

A Community Challenge to Benchmark Machine Learning Methods for Adaptive Immune Profiling

Chakravarthi Kanduri^{1,2}, Thomas Konstantinovskiy^{3,✉}, Puneet Rawat^{4,✉}, Milena Pavlovic^{1,2}, Damon H. May⁵, Rebecca Elyanow⁵, Bryan Howie⁵, Harlan S. Robins⁵, Crina Curca⁶, Bryan Hariadi⁶, Ashwath Kumar⁶, Jose Jacob⁶, Efthymia Papalexi⁶, Charles Roco⁶, Alex Rosenberg⁶, AIRR-Community Machine learning Working Group, Justin Barton⁷, Günter Klambauer⁸, Encarnita Mariotti-Ferrandiz⁹, Pieter Meysman¹⁰, Eline T. Luning Prak¹¹, Lindsay G. Cowell¹², Todd M. Brusko^{13,14,15}, Gur Yaari^{3,16,*}, Victor Greiff^{4,17,*}, Geir Kjetil Sandve^{1,2,*}

¹Scientific Computing and Machine Learning section, Department of Informatics, University of Oslo, Norway

²UiORealArt Convergence Environment, University of Oslo, Norway

³Faculty of Engineering and Bar Ilan Institute of Nanotechnology and Advanced Materials, Bar-Ilan University, Israel

⁴Department of Immunology, University of Oslo, Oslo, Norway

⁵Adaptive Biotechnologies - 1165 Eastlake Ave E, Seattle, WA 98109

⁶Parse Biosciences, Seattle, WA

⁷Institute of Structural and Molecular Biology, University of London, United Kingdom

⁸Institute for Machine Learning, Johannes Kepler University Linz, Linz, Austria

⁹Sorbonne Université, INSERM, UMRS959, Immunology-Immunopathology-Immunotherapy (i3) lab, Paris France (E.M.-F.).

¹⁰Adrem Data Lab, Department of Computer Science, University of Antwerp, Antwerp, Belgium

¹¹Department of Pathology and Laboratory Medicine, Perelman School of Medicine, University of Pennsylvania, Philadelphia

¹²Department of Health Data Science and Biostatistics, Peter O'Donnell Jr. School of Public Health; Department of Immunology, School of Biomedical Sciences; UT Southwestern Medical Center, Dallas, Texas, USA

¹³Department of Pathology, Immunology, and Laboratory Medicine, Diabetes Institute, College of Medicine, University of Florida, Gainesville, FL, USA

¹⁴Department of Pediatrics, College of Medicine, University of Florida, Gainesville, FL, USA

¹⁵Department of Biochemistry and Molecular Biology, College of Medicine, University of Florida, Gainesville, FL, USA

¹⁶Department of Pathology, Yale School of Medicine, New Haven, CT, USA

¹⁷Imprint Labs, LLC. New York, NY, USA

* Equal contribution, [✉] equal contribution

Correspondance: geirksa@ifi.uio.no, victor.greiff@medisin.uio.no, gur.yaari@yale.edu

Abstract

Adaptive immune receptors (B- and T-cell receptors; BCRs and TCRs) can recognise a diverse array of different antigens, initiating and maintaining the adaptive immune response. There is immense interest in academia and industry to discover immune receptors for diagnostics and therapeutics utilizing machine learning (ML). We propose the first community challenge to systematically benchmark ML methods on two tasks: given a dataset of immune state-labeled immune repertoires, predict (1) immune state (diagnostic use case), and (2) infer immune-state-associated receptors (therapeutic discovery use case). We conceived a benchmarking database of real-world relevant data including ~75,000 high-fidelity simulated (ground-truth) and experimentally generated TCR repertoires. By quantifying the performance of state-of-the-art techniques and identifying methodological gaps, this community challenge contributes to expediting ML-based solutions for immunodiagnostics and therapeutics discovery.

Introduction

Adaptive immune receptors (AIRs), comprising T-cell and B-cell receptors (TCRs and BCRs) are crucial for recognizing antigens and mounting an effective immune response. Genetic mutations that eliminate TCRs or BCRs lead to severe immunocompromise, highlighting their importance^{1,2}. AIR-antigen recognition is the first signal for activating B and T cells^{3,4}. Because AIRs can bind to many disease-associated antigens, they have significant potential as therapeutics and as diagnostic markers⁵⁻⁷. Following antigen exposure, specific immune receptors are clonally selected and expanded within the AIR repertoire (AIRR).^{8,9} Given the increasing availability of both naive and antigen-experienced AIRR datasets, there is significant interest in mining AIRR data to decipher antigen-specific immune responses to accelerate diagnostics and therapeutics discovery.

Inferring antigen-specific signatures remains challenging because of several factors (**Figure 1.a**). Technological limitations render the generation of large-scale data from antigen-binding and non-binding cells for many antigens impractical, resulting in limited public availability of such data. While immune repertoires labeled with immune states (e.g., health or disease) can be generated at scale, the antigen specificity of receptor sequences within repertoires remain unpredictable¹⁰. "Immune state" here refers to an individual's adaptive immune system response to an immune event (e.g., disease, vaccination, allergy). Additional challenges include (i) Cross-reactivity or multireactivity: immune receptors may bind molecularly similar antigens from multiple pathogens and self-antigens¹¹, (ii) Temporal changes: immune repertoires are dynamic and vary over time. (iii) High diversity and low overlap: vast V(D)J recombination diversity ($\sim 10^8$ – 10^9 unique immune receptor sequences^{12,13} per individual), combined with individual variation and minimal inter-individual overlap, complicates common pattern identification. Many immune responses are characterized by so-called private epitopes, though it is unclear if this is inherent or due to insufficient receptor cataloging in databases. Study design variations (e.g., sampling depth, number of individuals assayed, library preparation protocols) can also impact the observable pattern overlap across individuals¹⁴⁻¹⁷. Yet, similarities in immune receptors among patients with the same disease are increasingly being recognized¹⁸⁻²⁴. (iv) Incomplete knowledge of immune signals: identifying antigen-specific signatures in AIRR is a high-dimensional pattern detection problem with incomplete understanding of pattern structures (e.g., identical sequences²⁵⁻²⁷, similar sequences²⁷⁻³², shared subsequences³³⁻³⁶, sequence motifs³¹, clonal expansion³⁷)³⁸. We refer to these immune response-encoded changes in immune repertoires as immune signals³⁹. (v) Low signal abundance/witness rate: the prevalence of immune state-associated sequences within repertoires (referred to as signal abundance or witness rate in ML literature³¹) can occur as infrequently as 1 in 10^8 sequences in peripheral blood.

Traditional methods for identifying disease-associated changes in immune repertoires, including germline gene distribution, diversity metrics, and V(D)J recombination probabilities^{26,37,40} may capture a specific type of global repertoire-level changes in isolation, but fail to precisely identify potential disease-associated receptor-level sequence patterns embedded within repertoires. Over the past decade, there has been growing interest in leveraging the pattern recognition capacity of ML to discover disease-related features in AIR^{38,41,42}. This involves identifying patterns, such as shared public clones, similar sequences, subsequences, and sequence motifs, that recur across individuals. This problem resembles multiple instance learning (MIL)^{31,43} in the ML literature, where the labels of so-called bags (here, immune repertoires) are known, but the labels of instances of the bag (here, receptor sequences) are unknown. However, unlike typical MIL

problems with hundreds or thousands of instances, a sequenced immune repertoire contains up to a million instances sampled from a pool of approximately 10^9 instances⁴⁴ and, therefore, constitutes one of the most difficult MIL problems ever tackled. Additionally, the witness rate in a positive bag that indicates the correct class, e.g., the immune status, is extremely low for immune repertoires. Thus, we believe this represents a highly difficult yet promising challenge for ML^{31,45–47}. Current ML methods vary widely in their assumption, architecture, encoding, sample size etc. Although ML-based discovery of antigen-specific immune receptor features is considered a key technology for future precision medicine approaches⁴⁸, currently available ML methods have not been systematically benchmarked for their ability to recover disease-associated features and fundamental methodological advances are needed for computational immunology and ML.

ML progress in the AIRR field is hindered by a lack of large-scale data with so-called ground truth^{10,49}. ML textbooks and benchmarking literature define ground truth as prior knowledge on the data generation process and its characteristics such as labels of receptor sequences, forms of immune signals, and information about all the other (causal) factors used to generate the data^{49–51}. Large-scale ground truth data are required for the development and benchmarking of novel AIRR-adapted ML approaches⁵², but are often prohibitively expensive, impractical, or impossible (in some cases) to obtain experimentally. Most methods^{17,53–57} have been assessed only on insufficiently characterized experimental data that are limited in size and scope, hindering comprehensive benchmarking⁴⁹. Fortunately, simulation frameworks that accurately mimic the V(D)J-recombination process exist^{58–65} while preserving statistical properties of AIRR data⁵⁸ when introducing known ground truth signals^{66,67}. These tools have already been used for AIRR ML-based assessment studies^{57,67,68}.

We propose the first community challenge for ML-based adaptive immune profiling, designed to comprehensively assess ML methods across two tasks: **Task 1**: immune state prediction based on immune receptor repertoire (i.e., repertoire label prediction), and **Task 2**: identification of immune state-associated receptor sequences (i.e., biological interpretability). Task 1 quantifies method capacity to predict immune state as a function of “assessment attributes” (**Figure 1.d**) such as label-associated sequence patterns, signal abundance, and study design characteristics (e.g., dataset size, species richness, (number of unique clonotypes), signal abundance shifts, class imbalances, systematic differences between comparator groups, etc). Task 2 quantifies a given method’s capacity to identify immune state-determining (antigen-associated) receptors, also as a function of various assessment attributes mentioned above. Task 1 is relevant for AIRR-based diagnostics, while Task 2 is crucial for discovering antigen-specific AIRR-based immunotherapeutics (**Box 1** and **Figure 1.b**). Notably, some methods (e.g., MIL methods described above) approach Task 1 by first identifying immune state-determining receptors (Task 2).

To ensure a rigorous evaluation of state-of-the-art AIRR-ML methods across various complexities, benchmarking requires an extensive suite of datasets. However, resource-intensive model training on an extensive suite of datasets can lead to unnecessary computational expenditure. Therefore, the AIRR-ML challenge adopts two phases (**Figure 1.c**). Phase-1, an open Kaggle competition, identifies the top 10 ML methods across challenges. Phase-2 involves rigorous internal evaluation of top methods (in collaboration with method developers) on a broader benchmark suite, disentangling abilities across assessment attributes (**Figure 1.d**). For both phases of the evaluation, we will provide (a) public training datasets with ground truth immune signals (where relevant) and repertoire labels and (b) unlabeled testing datasets. Performance

will be quantified by the degree to which disease-associated immune signals were recovered, thus directly measuring relevance for discovering immune receptor-based diagnostics and therapeutics.

We conceived a benchmarking database comprising ~75,000 high-fidelity simulated (ground-truth) and experimental TCR repertoires, organized into ~100 distinct training (and corresponding test) datasets with approximately equal representation of synthetic and experimental datasets. Experimental datasets include one of the largest unpublished datasets for type 1 diabetes (T1D) and Herpes Simplex Virus 2 (HSV-2). These include n=1105 seropositive and n=1050 seronegative for T1D⁶⁹, and n=113 seropositive and 365 seronegative for HSV-2. Other published experimental repertoire datasets with known serostatus of Cytomegalovirus infection (CMV) and SARS-CoV-2 infection^{25,70} were also included. The synthetic datasets were generated using custom simulation frameworks^{66,67}.

We further generated pilot data to assess performance on both tasks across all datasets using two methods (one baseline, and one widely used AIRR-ML method²⁵). The pilot data address several key considerations: **a. Dataset suitability:** ensuring appropriate challenge levels. **b. Varying degrees of difficulty:** ensuring varying degrees of difficulty for the same assessment attribute. **c. Reference benchmarks of performance:** Providing a reference to participants with attainable performance levels, so that they can be improved upon. **d. Value of Sophisticated Methods:** Determining scenarios where advanced methods offer benefits over simpler approaches, ensuring innovation and progress through the community challenge.

While public challenges have driven ML innovation in technology and business (e.g., ImageNet, Netflix prize, MNIST database⁷¹⁻⁷³), such standardized challenges remain limited in life sciences (e.g., CASP, DREAM challenges^{74,75}) and are only emerging in AIR-based therapeutics discovery^{76,77}. We anticipate this community challenge will foster innovation and expedite ML approaches tailored to immune repertoire datasets, highlighting promising methods for AIRR-based diagnostics and therapeutics, and deepening our understanding of adaptive immunity.

The remainder of this work is structured into two main sections: **(i) Methods** and **(ii) Pilot data**. The pilot data section presents findings on data suitability and reference performance benchmarks. The methods section details: **(a) Phase-1: Competition-based identification of the best-performing AIRR-ML methods**, including Kaggle platform rationale, prize money prerequisites, dataset details, performance metrics, and scoring for leaderboard ranking. **(b) Phase-2: Stratified performance analysis of the top 10 ML methods across diverse characteristics**, covering the details of ~ 100 datasets, replication dataset design to assess performance variability, and data hosting/submission. **(c) Process for assessing data suitability and establishing reference benchmarks**, including details on the two ML methods and interpretability approaches used, and criteria for inclusion of benchmarking datasets based on pilot data.

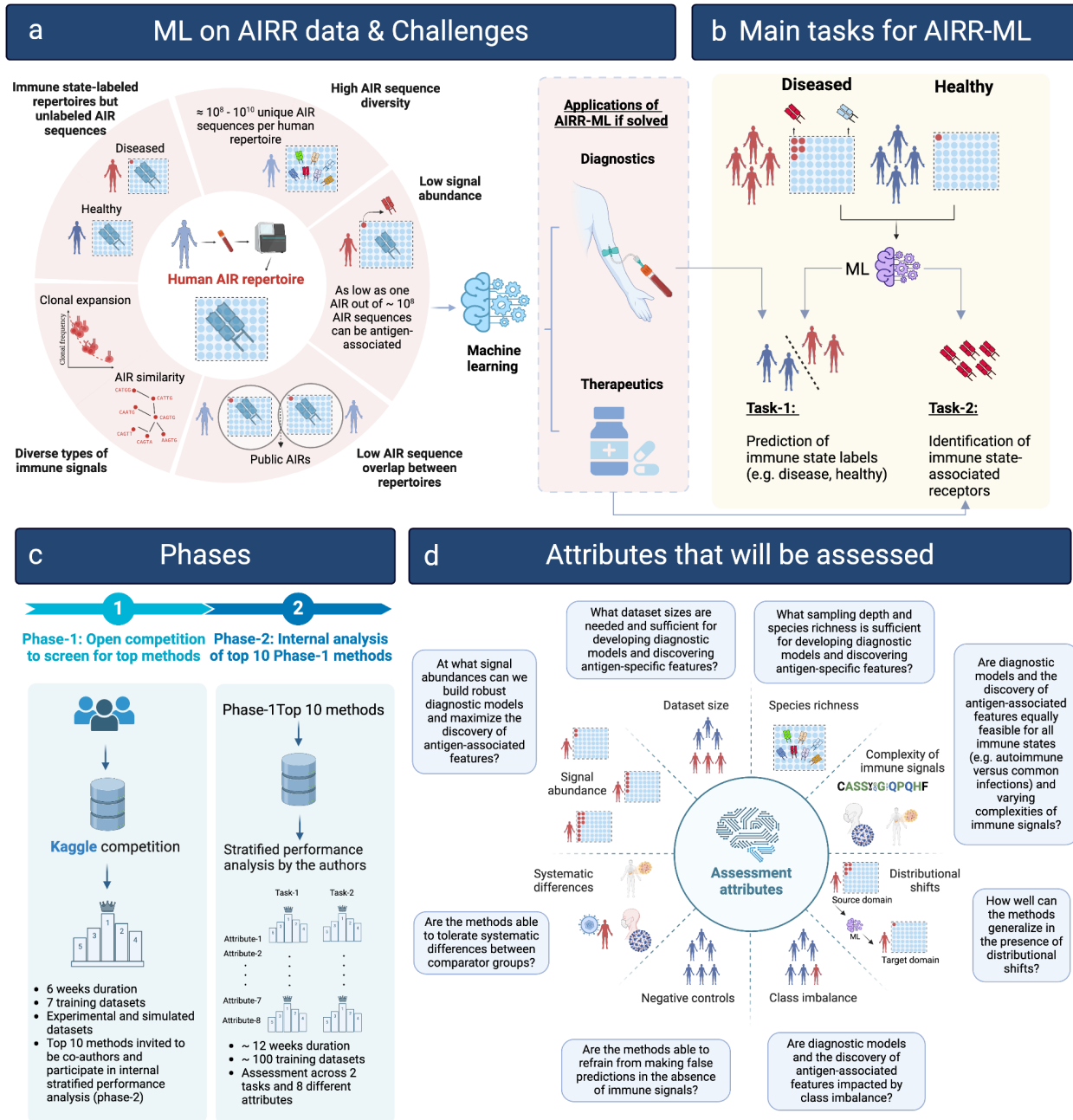


Figure 1: Overview of the AIRR-ML challenge. **a. ML on AIRR data and associated challenges.** ML on AIRR data has recently seen increased interest for its promise for diagnostics and therapeutics. However, implementing ML on AIRR data is not straightforward and presents several challenges: (i) Human AIR sequences have enormous diversity because of V(D)J recombination, and it is expected that there can be approximately 10^8 to 10^{10} unique TCRs/BCRs in an individual. (ii) Antigen-associated sequences are known to occur with a frequency as low as 1 out of 10^8 sequences. (iii) With current technological limitations, AIR sequences rarely have labels on which antigens they are associated with; only repertoire-level labels are available. (iv) Very little overlap ($< 1\%$) exists between the sampled AIR sequences of any two individuals. (v) The immune state-associated signals that explain the predictions of ML models have been detected in diverse forms, and their true form remains undeciphered and may be disease-dependent. **b. Main tasks for the AIRR-ML challenge.** The AIRR-ML challenge will evaluate the ML methods on the ability to accurately predict the immune states of repertoires (task-1) and the ability to identify the immune state-specific features (task-2). **c. Phases.** The evaluation will be carried out in two phases. Phase-1 involves an open Kaggle competition that will last for 6 weeks, where the participants are required to train models on seven distinct training datasets (both experimental and synthetic) and submit their repertoire label predictions and recovered immune signals on the test datasets. The leaderboard will

be based on a scoring scheme that considers both task-1 and task-2 but with more importance attributed to experimental data. The top 10 methods on the leaderboard will be invited to be co-authors on a manuscript that will report the findings of a comprehensive, stratified performance analysis. In Phase-2 of the evaluation, the best-performing methods will be trained on circa 100 training datasets across eight different tasks (shown in panel D) to perform a stratified analysis of the abilities of the methods as a function of different attributes of the datasets. **d. Assessment attributes.** The stratified evaluation of Phase-2 will involve assessing the abilities of the methods on eight different tasks that include varying dataset size and sampled species richness, increasing signal abundance and varying complexities of immune signals, varying degrees of distributional shift and class imbalance, etc.

Box 1: The AIRR-ML challenge overview

Task 1: Immune state prediction based on immune receptor repertoire

→ Goal: Quantify the capacity of community-sourced ML methods to predict the immune state (label) of a repertoire as a function of signal abundance, the complexity of immune state-associated features, dataset size, species richness, signal abundance shift, class imbalance, etc.

→ Real-world use: AIRR-based diagnosis of disease status

Task 2: Identification of immune-state-associated immune receptors

→ Goal: Quantify the capacity of community-sourced ML methods to identify immune state-determining (antigen-associated) receptors

→ Real-world use: Discovery of antigen-specific AIRR-based immunotherapeutics

Methods

Phase-1: Competition-based identification of the best-performing AIRR-ML methods

Choice of Kaggle as competition platform for Phase-1

We chose to use Kaggle⁷⁸ as a platform for the open competition in Phase-1 for two major reasons. First, Kaggle is a popular platform in the ML community for ML and data science competitions; thus, hosting a competition on Kaggle increases the chance of attracting a new pool of competitors beyond the existing method developers in the AIRR field. Second, Kaggle readily offers possibilities for (a) dataset hosting, (b) real-time leaderboards, (c) automated scoring using either preloaded or custom metrics, (d) discussion forums, and, most importantly, (e) computing resources for each competition participant in the form of Kaggle notebooks that are similar to the more familiar Jupyter notebooks. Kaggle notebooks are run in a remote computational environment. At the time of writing this report, Kaggle offers an execution time of 12 hours for CPU and GPU sessions and 9 hours for TPU sessions for each Kaggle notebook session. The technical specifications of the computational infrastructure that is offered are as follows: 20 Gigabytes of auto-saved disk space, additional scratchpad disk space that will not be saved, CPU (30 Gigabytes of RAM, 4 CPU cores), GPU (1 Nvidia Tesla p100 GPU with 4 CPU cores and 29 Gigabytes of RAM; 2 Nvidia Tesla T4 GPUs with 4 CPU cores and 29 Gigabytes of RAM each), TPU (P6 CPU cores with 330 Gigabytes of RAM). If the competition participants prefer to use their own computational infrastructure, they can download the Phase-1 competition datasets from Kaggle and return to Kaggle only to submit their predictions.

Prize Money for the Kaggle competition

A total prize pool of €10,000 will be distributed among the top three competitors of the Kaggle competition, with €5,000, €3,000, and €2,000 awarded for first, second, and third place, respectively. To win the prize money, a prerequisite is that the participants make their code open-source. Usage of covariate information supplied for the experimental datasets, such as HLA information, age, gender, etc., is allowed. Usage of pre-trained language models and other properties of amino acids (such as biophysicochemical properties and substitution matrices) is allowed, but note that the competition hosts will not provide these.

Datasets for the Kaggle competition

The Kaggle competition will consist of a total of eleven test datasets. These include five experimental datasets and six synthetic datasets that assume either public sequences or shared complex sequence patterns constitute the immune signals (**Figure 2.a**). Below is a detailed description of the datasets included in Phase-1. The datasets that will be used in Phase-2 for stratified performance analysis are discussed later in a subsection titled [Datasets for Phase-2](#).

Experimental data sources for the Kaggle competition

T1D cohort

We secured access to a large unpublished cohort of TCR β repertoires in T1D, consisting of ~ 1100 T1D and ~ 1100 non-T1D repertoires. A detailed investigation of this cohort is being considered for publication elsewhere⁶⁹. This T1D cohort is unique for a few reasons: (1) first, it represents one of the largest dataset of TCR β repertoires available to date and is yet unpublished. Being unpublished minimizes the chance of data leakage by competition participants' access to true labels. (2) second, the global repertoire-level changes in autoimmune conditions were found to be distinct and modest when compared to infections and cancer^{38,79,80}. In other words, the repertoire-level signatures in T1D can pose a different challenge to ML than what common infection datasets present. This allows the investigation of operating assumptions and architectural choices suited for distinct forms of immune states (common infection versus autoimmune). (3) Third, the availability of HLA allele information for the whole cohort, including T1D and non-T1D repertoires allows disentangling HLA-associated immune signals from T1D-associated immune signals⁸¹. (4) Fourth, this dataset is representative of most real-world experimental datasets in terms of biases (e.g., library preparation differences between batches, demographic variable overrepresentation such as age, sex, etc.). This allows a retrospective evaluation (after the conclusion of the challenge) of whether and how models are able to handle such biases.

Detailed metadata of each repertoire in the cohort, including age, sex, HLA genetic variation information, and the clinical status of individuals indicating whether one is T1D, a first-degree or second-degree relative of T1D, or unrelated controls without diabetes, is shown in **Table S1**. This information will be provided to the challenge participants using MiAIRR standard compliant field names⁸². In addition to the provided metadata, we do have knowledge of islet autoantibody information and disease duration. However, we refrain from providing this information since the same information is not available for those without T1D from the same cohort, which could allow the challenge participants to easily identify the true labels. The availability of such covariate information allows us to investigate (after the conclusion of the challenge) whether the top methods generalize well within the disease population stratified by different confounders.

Although nuances in clinical definition can lead to some degree of noise due to label misclassification (e.g., someone who is autoantibody-positive but does not have metabolic evidence of glucose dysregulation, first-degree relatives who can have high genetic risk), we intend to rely solely on the clinical definition of whether someone is diagnosed with T1D (T1D⁺) or not (T1D⁻) as two contrasting labels for the label prediction problem. The ultimate goal is to advance these efforts toward early-stage detection of T1D, prior to clinical diagnosis, using TCR biomarkers. **Table 1** shows the number of examples within training and each test dataset, as well as a summary of measures of age and sex. In ML literature “examples” typically refer to individual data points or observations. TCR β CDR3 junction amino acid sequences with V, D, and J gene information and duplicate counts of the sequences will be supplied for each repertoire using AIRR standards compliant⁸³ field names.

Kaggle allows limiting continuous leaderboard updates to a subset of examples. This prevents participants from over-relying on multiple submissions to overfit their solutions. (described in detail in sub-section [Continuous and final leaderboards](#)). **Table 1** also shows the number of examples used for continuous leaderboard updates. Note that test dataset s1_2 has the same set of instances as test dataset s1_1, but the instances will be weighted during evaluation to balance out age distribution differences between the two classes. Briefly, the probabilities of representation of each defined age group (relative frequencies) are computed for the entire dataset (training + test splits) separately for each class (T1D⁺ or T1D⁻). The inverse of these probabilities was assigned as weights to each instance in test dataset 2, depending on the instance’s age group and class, and will be used during evaluation for computing a weighted performance metric.

Table 1: Overview of experimental datasets included in the Kaggle competition.

Dataset	Cohort	Number of examples ^{\$}	Age in years (mean $\pm$ standard deviation)	Sex (female + male + missing data)	Used for continuous leaderboard updates?
Training dataset s1	T1D	908	27 $\pm$ 15	527 + 380 + 1	Not applicable
Test dataset s1_1	T1D	390 (~100*)	26 $\pm$ 15	239 + 151 + 0	Partially*
Test dataset s1_2 (Weighted instances $\square$)	T1D	390	26 $\pm$ 15	239 + 151 + 0	No
Test dataset s1_3	T1D	857	38 $\pm$ 19	449 + 403 + 5	No
Training dataset s2	HSV-2	302	38 $\pm$ 11	197 + 105 + 0	Not applicable
Test dataset s2_1	HSV-2	76	37 $\pm$ 11	47 + 29 + 0	Yes
Test dataset s2_2	HSV-2	100	40 $\pm$ 11	34 + 66 + 0	No

^{\$} In ML literature, “examples” typically refer to individual data points or observations. * Only the numbers in parentheses for test dataset s1_1 are used for continuous leaderboard updates. The remaining are used for the final scoring. $\square$ Weighted instances refer to the inverse probability weights assigned to each example to balance out age distribution differences as described in the main text of the T1D cohort description.

HSV-2 cohort

We secured access to another unpublished cohort of TCR β repertoires serologically labeled for HSV-2 infection, consisting of 113 examples that are seropositive and 365 examples that are seronegative. This cohort was provided by Adaptive Biotechnologies and was sequenced using their immunoSEQ assay version v4b. We split the cohort into a training and two separate test datasets. One of the test datasets will be used for the continuous leaderboard (as described above), and the other will be used for the final leaderboard. While the training dataset and the test dataset intended for the continuous leaderboard consist of examples with 100% self-reported White ancestry and predominantly female; the test dataset intended for the final leaderboard consists of examples with 100% self-reported Black ancestry and majority male. The rationale for separating train and one of the test datasets by ethnicity is to be able to analyze how the models perform within the same cohort used for training and also how they generalize to a different cohort. While sex is assumed to be of minimal importance to HSV-2 classification from the T-cell repertoire, ancestry acts as a proxy for HLA, ensuring meaningful differences between the train and second test datasets. HSV-2+ donors tended to be significantly older than HSV-2- donors. Since age has a profound effect on the T-cell repertoire that could be used to derive a false diagnostic signal, the HSV-2+ examples in each dataset were matched for age with HSV-2- examples using propensity score matching, with the poorest-matched control samples excluded. **Table 1** shows the number of examples within training and each test dataset, and includes summary statistics of age and sex. Table S2 shows detailed metadata of the cohort, including age, sex, ethnicity, and batch.

Synthetic data sources for the Kaggle competition

Below, we first describe the assumptions on immune signals in the synthetic datasets. Next, we introduce the simulation frameworks used and outline the default properties of synthetic datasets. Finally, we provide specific details about the synthetic datasets.

Assumptions on immune signals in the synthetic datasets

The synthetic datasets included in this study are designed to capture specific information encoded in repertoires in different types of immune states from previous research (**Figure 2.a**). Specifically, **(a) public sequences:** identical or similar AIR sequences are shared between individuals with a shared disease or phenotype (referred to as public sequences). Such public sequences can be observed with increased frequency in the immune state of interest relative to a control group, more than expected by chance^{25–27}. Individuals with a particular immune state may carry an excess of such immune state-associated public sequences. The incidence at which such immune state-associated public sequences can be observed in repertoires (signal abundance) may vary between different immune states and can depend on study design characteristics and other factors (e.g., number of sequences sampled per individual, enrichment strategies, HLA type, CDR3 length, etc.). **(b) Shared complex sequence patterns:** immune state-associated sequences may share short sub-sequence patterns (e.g., k-mers or similar), and such sequences may be observed with elevated frequency in an immune state of interest relative to a control group more than expected by chance^{34,84–87}. Again, the frequency of such observed immune state-associated public sequences can vary between different immune states and depend on the study design characteristics and other factors described above.

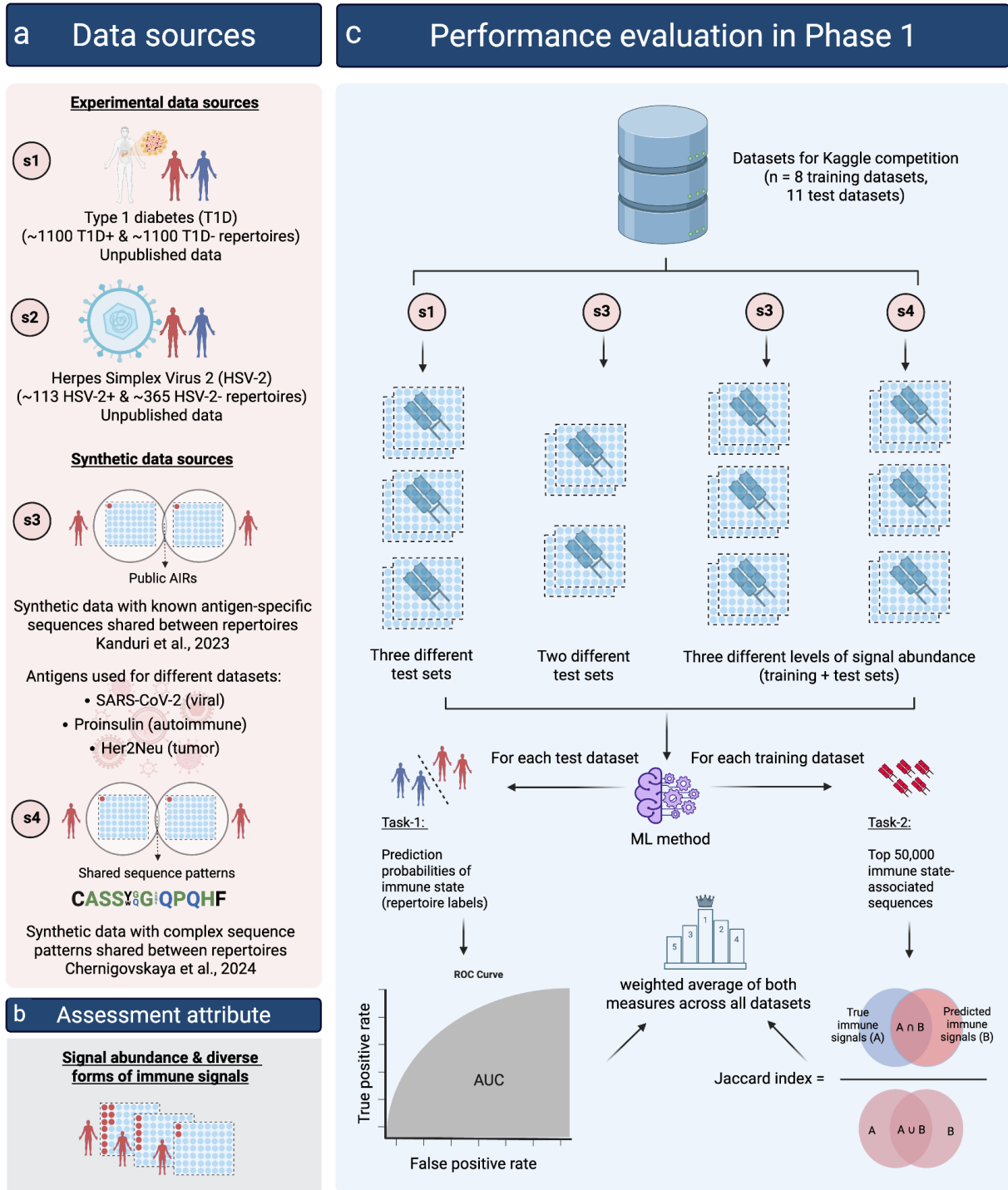


Figure 2: Overview of the datasets, assessment attributes and performance evaluation in Phase-1. **a. Data sources:** Four different data sources will be used in the Kaggle competition. This includes an unpublished experimental dataset consisting of T1D+ and T1D- repertoires (denoted as s1), an unpublished experimental dataset consisting of HSV-2+ and HSV-2- repertoires (denoted as s2), and synthetic datasets that either assume that public sequences or shared complex sequence patterns constitute the immune signals (denoted as s3 and s4, respectively). **b. Assessment attribute:** Phase-1 will solely focus on assessing the methods as a function of varying signal abundance and diverse forms of immune signals. **c. Performance evaluation in Phase-1:** The data sources described above (s1, s2, s3, s4) constitute the diverse forms of immune signals. For T1D (s1), one training dataset and three different test datasets are included. For HSV-2, one training and two different test datasets are included. For s3 and s4, three

different levels of signal abundance are included (training and corresponding test datasets). For each test dataset, the competition participants (depicted as one ML method here) must submit the prediction probabilities of the immune state (repertoire labels) and the top 50,000 immune state-associated sequences from the corresponding training dataset. These will be used to compute the performance metrics ROC AUC and Jaccard similarity, respectively, for each of the datasets. A weighted average of both measures across all the included datasets will be used as the basis for ranking on the leaderboard for the competition.

Simulation frameworks and default properties of synthetic datasets

All the synthetic datasets included in this study are simulated using the LIgO⁶⁷ and simAIRR⁶⁶ frameworks. The characteristics that can vary between the synthetic datasets are (a) background sequences used by the simulation frameworks for repertoire construction, (b) the source and number of immune state-associated sequences harbored by the positive-labeled repertoires, and (c) the total number of receptor sequences harboring an immune signal in a dataset. Note that background sequences here refer to sequences that are not associated with a particular immune event of interest in AIRR datasets. The exact details of the simulations, reproducible simulation specification files, and guidance on reproducing the simulations will be made available to the peer reviewers during the review stage but hidden from the competition participants (see [Code availability](#)). We refer to the following as the “default” properties of the synthetic datasets: A corresponding test dataset is included for all the synthetic training datasets included in the competition. Each training and test dataset consists of 400 repertoires, with balanced proportions of positive- and negative-labeled repertoires unless otherwise specified. Each repertoire comprises an average of 25,000 TCR β CDR3 junction amino acid sequences with V and J gene information. Nucleotide sequences and duplicate read count information are not included. While nucleotide sequences are omitted in the interest of file sizes of the entire benchmarking database, duplicate read count information is omitted from the synthetic datasets for the following reason: while the simulation of receptor sequence signals can also lean on insights from studies of antigen binding of individual receptors, there is no similar external basis to lean on regarding the clonal expansion of disease-relevant cells - in particular for the interaction between sequence signals and clonal expansion. To avoid any methods either exploiting unrealistically simulated clonal expansion patterns or getting thrown off by them, we chose to focus exclusively on receptor sequence signals in the synthetic benchmarking datasets. Furthermore, to avoid any systematic differences between positive and negative examples that some ML methods may exploit, we used the same data generation process for the construction of positive and negative examples, except for the subsequent introduction of immune signals into the positive examples.

Source of unpublished experimental antigen-specific sequences for simulations

We partnered with Parse Biosciences to acquire experimental antigen-specific TCR sequences for our simulations. Parse Biosciences' Evercode TCR mega assay allows for high-throughput, single-cell profiling of antigen-specific TCRs, analyzing up to one million cells simultaneously. Their workflow involves stimulating T cells with antigens and expanding them in vitro. From Parse Biosciences, we specifically obtained TCRs for eight different antigens spanning three classes: (a) autoimmune: glutamic acid decarboxylase 65 (GAD65), proinsulin, myelin basic protein (MBP), (b) viral: Epstein–Barr virus (EBV), SARS-CoV-2, Human papillomavirus (HPV), (c) tumor antigens: Melan-A (MART-1), Her2Neu. While this assay captures both alpha and beta chain CDR3s, we focused solely on beta CDR3s to align with the community challenge's requirements. A significant portion of these antigen-specific sequences are not publicly available. To prevent the shortcut of identifying potential immune state-associated sequences through public database mining, we filtered out any sequences already present in public databases before

using them in our simulations. More detailed information on how these unpublished sequences were used for different datasets can be found in the subsequent sections on synthetic data generation.

Synthetic datasets for Phase-1

The synthetic datasets in the Kaggle competition are designed to assess the impact of signal abundance and diverse forms of signal patterns on the performance of ML methods in both repertoire label prediction and recovery of immune signals (**Figure 2.b**). Signal abundance here refers to the percentage of unique sequences within a repertoire that are immune state-associated. The different immune signal patterns of the synthetic data sources (**Figure 2.a**) constitute diverse forms of immune signals. **Table 2** shows precise details of the characteristics that vary between the six synthetic datasets included in the Kaggle competition. Using simAIRR⁶⁶, we generated three datasets with increasing signal abundance (5, 20, and 100 immune state-associated sequences per positive-labeled repertoires), where shared public sequences constitute the immune signals. In each dataset, known antigen-specific experimental sequences acquired from Parse Biosciences were used as immune state-associated features. For this, we used sequences specific to Proinsulin (autoimmune), SARS-CoV-2 (viral), Her2Neu (tumor antigen). A random sample of experimental sequences from the ImmuneCode database⁷⁰ were used as the background sequences for repertoire construction. Using LIgO³⁹, we generated three datasets with increasing signal abundance (5, 20, and 100 immune state-associated sequences per positive labeled repertoires), where shared complex sequence patterns constitute the immune signals. Specifically, the immune state-associated sequences share short sub-sequence patterns spanning 4 to 6 amino acid residues with up to a maximum of 3 gapped positions obtained through rejection sampling of sequences from known V(D)J recombination models^{58,88}. A random sample of sequences from known V(D)J recombination models were used as background sequences to construct the repertoires^{58,88}.

Performance Metrics and Leaderboard in the Kaggle competition (Phase-1)

Since the AIRR-ML challenge involves the assessment of both repertoire label prediction accuracy (Task 1) and interpretability (Task 2) across multiple datasets, a single performance metric that combines both aspects across all datasets is needed for the Kaggle competition leaderboard (see **Figure 2.c**). The individual performance metrics for the Kaggle competition that will be combined into a single score are described below. The details of how they will be combined and what the leaderboard will be based on are described in a later subsection titled Leaderboard in the Kaggle competition (Phase-1).

Quantification of classification performance using ROC AUC

The widely used receiver operating characteristic (ROC) curve is a visual representation that assesses the performance of a model across various possible thresholds of the prediction probabilities. The area under the curve (AUC) of the ROC⁸⁹ represents a single number that summarizes the model's performance across all the possible thresholds assessed and serves as a useful metric to compare models as long as the datasets are roughly balanced. Taking into account the probabilities of class assignments is one of the main reasons for preferring ROC AUC over other metrics to be reported as the main metric. The competition participants will be asked to return prediction probabilities for the immune state of each repertoire in each test dataset, enabling the computation of ROC AUC.

Table 2: Characteristics that vary between the six synthetic datasets included in the Kaggle competition.

Dataset	Immune state -associated features	Background sequences* used by simulation frameworks ^{66,67} for repertoire construction	Number of immune state-associated sequences per repertoire in positive-labeled repertoires	Total number of distinct immune state-associated sequences in the dataset
1	Proinsulin-specific sequences from Parse Biosciences	A random sample of experimental TCR sequences	5	650
2	SARS-CoV-2-specific sequences from Parse Biosciences	A random sample of experimental TCR sequences	20	940
3	Her2Neu-specific sequences from Parse Biosciences	A random sample of experimental TCR sequences	100	1720
4	Sequences sharing short sub-sequence patterns spanning 4 to 6 amino acid residues with up to a maximum of 3 gapped positions	A random sample of synthetic TCR sequences drawn from known V(D)J recombination models	5	2000
5	Sequences sharing short sub-sequence patterns spanning 4 to 6 amino acid residues with up to a maximum of 3 gapped positions	A random sample of synthetic TCR sequences drawn from known V(D)J recombination models	20	8000
6	Sequences sharing short sub-sequence patterns spanning 4 to 6 amino acid residues with up to a maximum of 3 gapped positions	A random sample of synthetic TCR sequences drawn from known V(D)J recombination models	100	40000

* Background sequences in the context of synthetic AIRR datasets refer to sequences that are not associated with a particular immune event of interest. E.g., when analysing T1D+ repertoire, all the other sequences that are not T1D-associated.

Quantification of recovery of immune signals using Jaccard Similarity

The Jaccard similarity⁹⁰ measures the similarity between finite sample sets and is computed as the ratio of the size of the intersection to the size of the union. By penalizing both false positives and false negatives, Jaccard similarity acts as a single intuitive metric that captures the trade-off between precision and recall⁹⁰ and is thus preferred for the Kaggle competition for evaluating the ability of ML methods to recover the true immune signals. Note that throughout the manuscript, “true immune signals” refer to the immune state-associated sequences used for simulations in synthetic data, where prior knowledge of receptor labels and the nature of sequence patterns exist from the outset. The competition participants will be asked to return a ranked list of the top 50,000 unique rows (including junction amino acid sequence, V, and J genes) that best contribute to the optimal classification for each training dataset, irrespective of the data encoding or representation used. We chose to base the predicted immune signals on training data, rather than test data,

to ensure that methods could recover the full set of true immune signals in all cases, as test data alone may contain only a subset of the signals in some cases. Although this number (50,000 predicted immune receptor sequences) exceeds the number of true immune state-associated sequences in the datasets, the number of sequences that will be used in the computation of the similarity index will be matched with the number of true signals (**n**) for each dataset by selecting the top “**n**” sequences. Independent of the type of information encoded in the repertoires as immune signals (public sequences or enriched sub-sequence patterns), the rows from the examples (repertoires) that harbor the immune signals will alone be used in the measurement of Jaccard similarity. The similarity between the predicted immune signals of the ML methods and the true immune signals will be measured as follows:

$$J(A, B) = \frac{A \cap B}{A \cup B}$$

where set A represents the set of predicted immune signals by the methods and set B represents the set of true immune signals. Note that “interpretability” is not applicable in the case of experimental datasets, as the knowledge of true immune state-associated sequences is lacking.

Leaderboard in the Kaggle competition (Phase-1)

The leaderboard in the Kaggle competition will be based on a single score that combines both ROC AUC and Jaccard similarity across all datasets (training + test combination). For this, we will use a weighted average of both the measures across all datasets as a single score for ranking on the leaderboard (**Figure 2.c**). The reason for choosing a weighted average is as follows: there are three experimental test sets and six synthetic test sets in Phase-1. Although fewer test sets in number, we intend to give equal importance to experimental data in the final leaderboard score. For this, the weights are calculated in such a way that experimental data always accounts for 50% of the leaderboard score.

Continuous and final leaderboards

There will be two different leaderboards: **(a)** a continuous leaderboard that will be used throughout the duration of the competition and **(b)** a final leaderboard that will be revealed only at the conclusion of the competition. The reason for having two different leaderboards is as follows: since the Kaggle platform allows multiple submissions per day for each competition participant, getting continuous feedback on all the test data means that the leaderboard score can be optimized by making many submissions. To avoid such hacks, a common practice (that Kaggle also provides) is to set aside a subset of the test data that will never be used when giving continuous feedback throughout the competition and will only be used at the conclusion of the competition for the final ranking. The ROC AUC on two of the three experimental test sets and the Jaccard similarity of all the test sets will be excluded from the continuous leaderboard. In contrast, the final leaderboard will be based on a weighted average of both the classification performance and interpretability of all nine test datasets. **Table 3** shows which test datasets and evaluation criteria will be used for the continuous leaderboard versus the final leaderboard. Because of using different test datasets and evaluation criteria for the two different leaderboards, the weights that will have to be used for the weighted average (described above) also differ and are shown separately in **Table 3**. As “interpretability” is not applicable in the case of experimental datasets, the Jaccard similarity of experimental test sets is assigned a weight of zero as shown in **Table 3**.

The weighted average that will be used for the continuous and final leaderboards will be as follows, where w_i and W_i represent the weights that will be used for the continuous and final leaderboard, respectively, T_i represents the respective test sets, AUC represents the ROC AUC and JS represents the Jaccard similarity measure.

$$\text{continuous leaderboard score} = \frac{w_1 AUC_{T1} + w_2 AUC_{T2} + w_3 AUC_{T3} + \dots + w_{21} JS_{T10} + w_{22} JS_{T11}}{w_1 + w_2 + w_3 + \dots + w_{20} + w_{21} + w_{22}}$$

$$\text{final leaderboard score} = \frac{W_1 AUC_{T1} + W_2 AUC_{T2} + W_3 AUC_{T3} + \dots + W_{21} JS_{T10} + W_{22} JS_{T11}}{W_1 + W_2 + W_3 + \dots + W_{20} + W_{21} + W_{22}}$$

As an example, if an ML method attains 0.8 as ROC AUC across all test sets for repertoire label prediction and 0.4 as Jaccard similarity across all the synthetic test sets, then, according to the weights shown in **Table 3**, the final leaderboard score would be 0.7 (16.8/24).

Table 3: Relative importance given to the repertoire label prediction accuracy and interpretability of the models on different test datasets.

Training dataset	Corresponding test set	Weights for each ROC AUC score towards the continuous leaderboard	Weights for each ROC AUC score towards the final leaderboard	Weights for each Jaccard similarity score towards the continuous leaderboard	Weights for each Jaccard similarity score towards the final leaderboard
Experimental dataset S1	Experimental test set 1 (T1)	5 (w_1)	2 (W_1)	0 (w_{12})	0 (W_{12})
Experimental dataset S1	Experimental test set 2 (T2)	0 (w_2)	3 (W_2)	0 (w_{13})	0 (W_{13})
Experimental dataset S1	Experimental test set 3 (T3)	0 (w_3)	3 (W_3)	0 (w_{14})	0 (W_{14})
Experimental dataset S2	Experimental test set 4 (T4)	1 (w_4)	1 (W_4)	0 (w_{15})	0 (W_{15})
Experimental dataset S2	Experimental test set 5 (T5)	0 (w_5)	3 (W_5)	0 (w_{16})	0 (W_{16})
Synthetic dataset 1	Synthetic test set 1 (T6)	1 (w_6)	1 (W_6)	0 (w_{17})	1 (W_{17})
Synthetic dataset 2	Synthetic test set 2 (T7)	1 (w_7)	1 (W_7)	0 (w_{18})	1 (W_{18})
Synthetic dataset 3	Synthetic test set 3 (T8)	1 (w_8)	1 (W_8)	0 (w_{19})	1 (W_{19})
Synthetic dataset 4	Synthetic test set 4 (T9)	1 (w_9)	1 (W_9)	0 (w_{20})	1 (W_{20})
Synthetic dataset 5	Synthetic test set 5 (T10)	1 (w_{10})	1 (W_{10})	0 (w_{21})	1 (W_{21})
Synthetic dataset 6	Synthetic test set 6 (T11)	1 (w_{11})	1 (W_{11})	0 (w_{22})	1 (W_{22})

w_i and W_i represent weights that will be used for continuous and final leaderboards, respectively, and T_i represents the respective test sets.

Accommodating multiple test sets and evaluation criteria in the Kaggle platform

The Kaggle platform typically allows only a single predictions file per participant (test.csv) and a true solutions file from the competition hosts (solutions.csv). To evaluate multiple test datasets and metrics

related to label prediction and interpretability, we will implement a custom evaluation metric on the platform. Here, the custom metric refers to the continuous and final leaderboard score described above. Each repertoire in the test datasets is assigned a unique identifier. The solutions.csv file will include five columns: (1) The “ID” column will link the submitted results with the true solutions. (2) The “label” column will contain True/False values for each repertoire in the test set. (3) The “test_dataset” column indicates which respective test dataset the solutions correspond to. (4) The “public” column, which will also be boolean (True/False), determines whether a particular sample contributes to the continuous leaderboard or the final one. (5) The “signal” column will store a delimiter-separated string (of junction amino acid sequence, V and J gene information) representing the true immune signals (one sequence per row) for each dataset. In other words, for each test dataset, the solutions.csv contains as many rows as there are examples in the dataset + as many true immune state-associated sequences the corresponding training dataset has. In the test.csv, the participants will be required to include “ID”, “test_dataset”, “probability”, and “signal” columns, where probability refers to the label prediction probabilities, and the “signal” column contains the top 50,000 predicted immune signals. For each test dataset, the number of rows in test.csv will be based on the number of examples + 50,000. See [Table S3](#) for examples of solutions.csv and test.csv.

Phase-2: Stratified performance analysis of the top 10 ML methods across diverse characteristics

Based on the final leaderboard of the Kaggle competition in Phase 1, the top 10 methods (contestants) will be invited to become co-authors of the manuscript that reports a comprehensive, stratified investigation of the performance according to particular individual attributes of the datasets. As part of such an effort, the participants are required to contribute predictions to the full benchmarking database. They will be offered to do this in either of two ways: either by receiving the full database and returning their full suite of predictions or by providing us with a Docker container of their method that adheres to a uniform interface for running it (given that computational needs are also within the bounds of what we can ourselves handle – which we limit to 50 000 CPU hours and 500 GPU hours on a GPU at the level of NVIDIA Tesla V100). To estimate the uncertainty of performance, three independent datasets will be provided, each replicating each characteristic under investigation. If participants choose to train the models on their own computational resources, they will be allowed to return predictions either for the full database, including replicate datasets, or for only one dataset in each setting. This is to cater to the varying availability of computational resources.

Datasets for Phase-2

Below, we first describe the experimental and synthetic data sources that form the basis for the construction of benchmarking datasets in Phase 2 (**Figure 3.a**). We will then provide an overview of the benchmarking datasets (both experimental and synthetic) relevant to each assessment attribute.

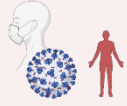
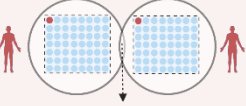
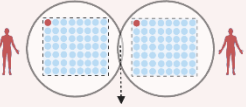
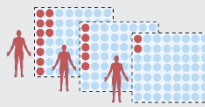
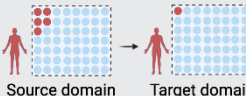
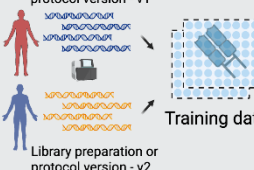
a Data sources		b Datasets for Phase 2			
Experimental data sources s1  SARS-CoV-2 infection (~1400 SARS-CoV-2+ repertoires) Public data - Nolan et al., 2020 s2  Type 1 diabetes (T1D) (~1100 T1D+ & ~1100 T1D- repertoires) Unpublished data s3  Cytomegalovirus (CMV) infection (~300 CMV+ & ~300 CMV- repertoires) Public data - Emerson et al., 2017 Synthetic data sources s4  Public AIRs Synthetic data with known antigen-specific sequences shared between repertoires Kanduri et al., 2023 Antigens used for different datasets: • Glutamic acid decarboxylase 65 (GAD65) • Proinsulin • Myelin basic protein (MBP) • Melan-A (MART-1) • Human papillomavirus (HPV) • Epstein-Barr virus (EBV) • Cytomegalovirus (CMV) • Influenza A s5  Shared sequence patterns CASSYGIQPHF Synthetic data with complex sequence patterns shared between repertoires Chernigovskaya et al., 2024		Assessment attributes Signal abundance & diverse forms of immune signals  Training dataset size  Species richness  Signal abundance shift  Source domain Target domain Systematic difference between examples  Library preparation or protocol version - v1 Library preparation or protocol version - v2 Training data Class imbalance  Negative controls 	Number of training datasets (experimental + synthetic) 20 (2 + 18) 27 (18 + 9) 27 (18 + 9) 6 (0 + 6) 10 (10 + 0) 6 (0 + 6) 2 (1 + 1)	Parameter configurations being assessed Signal abundance rates: 0.02%, 0.08%, 0.4% 100, 200, and 400 training examples 6000, 12000, and 25000 unique sequences Training 0.16% → test 0.08%; training 0.08% → test 0.02% Ten different training and test datasets with systematic differences between examples Positive label rate in training: 20%, 50%, 80% Two datasets with shuffled labels	Dataset sources s2 - s5 s2 - s4 s2 - s4 s4 s1 - s3 s4 s2, s4

Figure 3: Overview of the data sources, assessment attributes and datasets of Phase-2. **a. Data sources:** Five different data sources will be used in Phase-2. This includes three experimental data sources consisting of SARS-CoV-2+ repertoires (denoted as s1), T1D+ and T1D- repertoires (denoted as s2), and CMV+ and CMV- repertoires (denoted as s3). The synthetic data sources include datasets that either assume that public sequences or shared complex sequence patterns constitute the immune signals (denoted as s4 and s5, respectively). **b. Datasets for Phase-2:** The four different columns show the distinct assessment attributes, the total number of datasets (experimental and synthetic) that will be used for a particular assessment attribute, the parameter configurations being assessed for that assessment attribute and the corresponding data sources used for constructing the benchmarking datasets.

Experimental data sources for stratified performance analysis

Three different experimental data sources will be used in Phase 2 for a stratified performance analysis of the different attributes (see s1-s3 in **Figure 3.a**). One of the data sources is the unpublished T1D dataset that has been described above under the section "Datasets for the Kaggle competition". The other two data sources are published and available through public databases, as detailed below.

CMV TCR dataset

The CMV dataset has been published and described extensively elsewhere²⁵, and is available through AIRR Data Commons^{91,92}. Briefly, this dataset contains T-cell repertoires of circa 700 subjects with known serostatus of CMV infection. Different sets of independent and randomly selected repertoires from this dataset will be used across the assessments in Phase 2.

SARS-CoV-2 TCR dataset

The SARS-CoV-2 dataset is publicly available and has been described extensively elsewhere⁷⁰, and is available through AIRR Data Commons^{91,92}. This dataset contains T cell repertoires of circa 1400 subjects exposed to or infected by SARS-CoV-2. Note that this dataset does not contain repertoires of unexposed or uninfected individuals. Different sets of independent and randomly selected repertoires from this dataset will be used across the assessments in Phase 2.

Synthetic data sources for stratified performance analysis

The data sources that will be used in Phase-2 for the construction of synthetic benchmarking datasets are the same as those used in Phase-1 for the Kaggle competition (see s4 and s5 in **Figure 3.a**). Briefly, what differs between the two different synthetic data sources are the assumptions regarding what constitutes immune signals: either public sequences or shared complex sequence patterns. These general assumptions regarding immune signals in synthetic datasets drawn from prior literature have been described in detail above under Phase-1 in the section titled "Assumptions on immune signals in the synthetic datasets".

The simulation frameworks used to generate the synthetic datasets and the default characteristics of synthetic datasets in Phase-2 also remain unchanged from Phase-1. These have been described in detail above under Phase-1 in the section titled "Simulation frameworks and default properties of synthetic datasets". Note that reproducible simulation specification files with exact details of the simulations together with guidance on how to reproduce the simulations will be made available to peer reviewers during the review stage as described above (see [Code availability](#)). **Supplementary tables S4-S10** provide specific details of the simulation parameters for all the diverse scenarios that will be assessed in the stratified performance analysis.

Benchmarking datasets relevant to each assessment attribute

To rigorously evaluate the behavior and robustness of ML methods under varying conditions, the subsequent phase of stratified performance analysis requires a large benchmarking database. For this, we designed and constructed approximately 100 distinct training datasets (and corresponding test datasets) representing diverse scenarios, where 50% of the datasets originated from published/unpublished experimental data sources, while 50% of the datasets were synthetic. The datasets were designed to evaluate the performance of ML methods on both repertoire label prediction and interpretability as a function of multiple assessment attributes (**Figure 3.b**). The evaluation across different assessment attributes is aimed at addressing the broad questions described in **Box 2**. Below, we first introduce how replication datasets are designed to assess variability of model performance. We then provide an overview of datasets relevant to each assessment attribute.

Replication datasets to assess variability of model performance

To minimize the possibility of methods exploiting any particular simulation-specific instances, Phase-2 will include additional “replication” datasets for each attribute being assessed. For experimental datasets, the “replication” refers to independent repertoire datasets that share cohort characteristics (e.g., immune status and demographic variables) similar to those of the original experiments but might not necessarily share similar initial experimental conditions like library preparation methods. In the case of simulated datasets, replicates refer to independently generated datasets obtained by repeating the simulation process with different random seeds or initial conditions to obtain a distinct set of random data points and signals.

Datasets assessing the impact of signal abundance and diverse form of immune signals on repertoire label prediction and recovery of immune signals

A total of 20 datasets (2 experimental and 18 synthetic) are included for assessing the impact of signal abundance and diverse signal patterns on repertoire label prediction and recovery of immune signals (**Figure 3.b**). Training datasets include 400 examples with balanced labels and test datasets include 400 examples except for CMV-based datasets where only 200 examples are included in the test data owing to the available cohort size. These dataset sizes are applicable for all the datasets included in Phase-2 unless otherwise specified. The distinct immune states in experimental datasets (autoimmune, common infection), and the distinct forms of sequence patterns in synthetic datasets constitute diverse forms of immune signals. For synthetic datasets, using simAIRR⁶⁶, we generated a total of nine datasets with three different levels of signal abundance (0.02%, 0.08% and 0.4%), each replicated three times, where shared public sequences constitute the immune signals. In each dataset, known antigen-specific experimental sequences acquired from Parse Biosciences were used as immune state-associated features. For this, we used sequences specific to Proinsulin, GAD65 (autoimmune), EBV (viral), MART-1 (tumor antigen). Note that there was no overlap of immune signals between the synthetic datasets, meaning that different sets of sequences are used as immune signals for each of the nine datasets. A random sample of experimental sequences from ImmuneCode database⁷⁰ were used as the background sequences for repertoire construction. Using LIgO⁶⁷, we generated a total of nine datasets with three different levels of signal abundance (0.02%, 0.08% and 0.4%), each replicated three times, where shared complex sequence patterns constitute the immune signals. Specifically, the immune state-associated sequences share short sub-sequence patterns spanning 4 to 6 amino acid residues with up to a maximum of 3 gapped positions that were obtained through rejection sampling of sequences from known V(D)J recombination models^{58,88}. A random sample of sequences from

known V(D)J recombination models were used as background sequences to construct the repertoires^{58,88}.
Table S4 shows precise details of the characteristics that vary between the datasets.

Box 2: Questions to be addressed in the stratified performance analysis (Phase 2)

How well can the methods perform on Task 1 and Task 2 (see Box 1):

Q1. with increasing signal abundance and varying complexity of immune signals:

- Are diagnostic models and the discovery of immune state-associated features equally feasible for all immune states (e.g., autoimmune versus common infections)? At what signal abundances can we build robust diagnostic models and maximize the discovery of immune state-associated features? (Datasets described in **Table S4** and pilot data presented in **Figure 4**).

Q2. with varying dataset sizes:

- Are the current training dataset sizes of AIRR-ML studies sufficient for developing well-generalizable diagnostic models and discovering immune state-associated features? (Datasets described in **Table S5** and pilot data presented in **Figure 5**).

Q3. with varying sampling depth and resulting species richness:

- What sampling depth, and resulting species richness, is sufficient for developing generalizable diagnostic models and discovering immune state-associated features. (Datasets described in **Table S6** and pilot data presented in **Figure 5**).

Q4. in the presence of signal abundance shifts:

- Signal abundance shift refers to the difference in the signal abundance between the training dataset and the future datasets met by the trained model. At what level of signal abundance shift do the models start losing generalizability and what current strategies of the state-of-the-art methods make models adapt well to such shifts? (Datasets described in **Table S7** and pilot data presented in **Figure 6**).

Q5. in the presence of class imbalances:

- Is it better to have more positive examples than negative examples or vice versa and how does the learning vary based on the nature of imbalance? What current strategies of the state-of-the-art methods make models adapt well to class imbalance? (Datasets described in **Table S8** and pilot data presented in **Figure 6**).

Q6. in the presence of systematic differences between comparator groups:

- What is the impact of choosing control repertoires⁹³ from a different study (e.g. with different wet or dry bench protocols, different cell subsets, anatomic compartments or different immune states) on ML model performance? Do the models learn features associated with the immune state of interest (and are therefore generalizable) or do they learn the systematic differences between the comparator groups? (Datasets described in **Table S9** and pilot data presented in **Figure 7**).

In addition, the following questions will be addressed in the analysis of **negative controls**⁹⁴:

Q7. Negative controls: In “negative control” experiments, the treatment being investigated is absent, and therefore, should not show any changes in the experiment. In the absence of immune signals, the performance of any ML model should be close to random. Is that always the case, or do the methods still learn some dataset-specific biases? (Datasets described in **Table S10** and pilot data presented in **Figure 4**).

Datasets assessing the impact of training dataset sizes on repertoire label prediction and recovery of immune signals

A total of 27 datasets (18 experimental and 9 synthetic) are included for assessing the impact of training dataset sizes on repertoire label prediction and recovery of immune signals (**Figure 3.b**). The training dataset size is varied as 100, 200 or 400 training examples with balanced labels. Test datasets always included 400 examples, except for the CMV-based datasets, as stated above. The data sources for experimental datasets include the T1D-based cohort and the CMV-based cohort. At each training dataset size, three replicated datasets were constructed for both the experimental and the synthetic datasets. In the experimental datasets, a random sample of the desired number of examples from the original cohort was used for constructing the replicated datasets. For synthetic datasets, using simAIRR ⁶⁶, we generated a total of nine datasets with three different levels of training dataset sizes as stated above, each replicated three times, where shared public sequences constitute the immune signals. The signal abundance rate was maintained at 0.08%. In each dataset, known antigen-specific experimental sequences acquired from Parse Biosciences were used as immune state-associated features. For this, we used sequences specific to Proinsulin, MBP (autoimmune), HPV, EBV (viral), MART-1 (tumor antigen). Note that there was no overlap of immune signals between the synthetic datasets, meaning that different sets of sequences were used as immune signals for each of the nine datasets. A random sample of experimental sequences from the ImmuneCode database ⁷⁰ were used as background sequences for repertoire construction. **Table S5** displays details of the characteristics that vary between the datasets.

Datasets assessing the impact of species richness on repertoire label prediction and recovery of immune signals

A total of 27 datasets (18 experimental and 9 synthetic) are included for assessing the impact of species richness on repertoire label prediction and recovery of immune signals (**Figure 3.b**). Species richness was varied as 6000, 12000, or 25000 unique sequences on average per repertoire, both in training and test datasets. The data sources for experimental datasets include T1D-based cohort and CMV-based cohort. At each species richness, three replicated datasets were constructed for both experimental and synthetic datasets. In experimental datasets, a random sample of the desired number of examples from the original cohort was used for constructing replicated datasets. To mimic lower species richness in experimental repertoires, a sub-sampling procedure was implemented as described in the next subsection. For synthetic datasets, using simAIRR ⁶⁶, we generated a total of nine datasets with three different levels of species richness stated above, each replicated three times, where shared public sequences constitute the immune signals. The signal abundance rate was maintained at 0.16%. In each dataset, known antigen-specific experimental sequences from VDJdb database ⁹⁵ were used as immune state-associated features. For this, we used sequences specific to Influenza A. As above, there was no overlap of immune signals between the synthetic datasets, meaning that different sets of sequences were used as immune signals for each of the nine datasets. A random sample of experimental sequences from ImmuneCode database ⁷⁰ were used as the background sequences for repertoire construction. **Table S6** shows details of the characteristics that vary between the datasets.

Subsampling of sequences to mimic lower species richness in experimental repertoires

To subsample an experimental repertoire to mimic lower species richness, the repertoire is considered as a bag where some sequences are observed only once while some other sequences are observed more than once (typically represented by the `duplicate_count` of sequences in AIRR schema ⁹⁶). The sum of duplicate

counts in a repertoire is divided by the total number of unique sequences in the repertoire to obtain a duplicate count factor. The sum of duplicate counts in the subsampled repertoire (**c_sum_subsampled**) will be a product of the total number of desired unique sequences in the repertoire and the duplicate count factor, rounded to the nearest integer. A total of **c_sum_subsampled** sequences will be drawn randomly from the expanded bag of sequences of the repertoire (where some sequences are observed multiple times). The frequency with which a sequence is observed in the sub-sampled sequences will be the **duplicate_count** of that sequence. In Python, this is achieved by weighted random sampling of sequences using `numpy.random.choice`, where the relative frequencies of sequences ($\text{duplicate_count}/\text{sum}(\text{duplicate_counts})$) are used as weights and a total of **c_sum_subsampled** sequences are drawn (and where multiple draws of the same sequence are combined to a single sequence entry with duplicate count reflecting the number of times that sequence was drawn). If the total number of desired unique sequences in the subsampled repertoire is not met with the above described procedure, the number of desired unique sequences is modulated until the desired richness is achieved without disturbing the duplicate count factor.

Datasets for assessing the impact of signal abundance shifts on repertoire label prediction and recovery of immune signals

A total of 6 synthetic datasets were included for assessing the impact of signal abundance shifts on repertoire label prediction and recovery of immune signals (**Figure 3.b**). Specifically, we included two different variations of signal abundance shifts, where in one variation, the signal abundance dropped from 0.16% in the training dataset to 0.08% in the test dataset, and in another variation, it dropped from 0.08% in the training dataset to 0.02% in the test dataset. Three replicated datasets were constructed for each of the variations. Using `simAIRR`⁶⁶, we generated a total of six datasets, where shared public sequences constitute the immune signals. In each dataset, known antigen-specific experimental sequences from VDJdb database⁹⁵ were used as immune state-associated features. For this, we used sequences specific to EBV. Again, there was no overlap of immune signals between the datasets, meaning that different sets of sequences were used as immune signals for each of the six datasets. A random sample of experimental sequences from ImmuneCode database⁷⁰ were used as the background sequences for repertoire construction. **Table S7** shows details of the characteristics that vary between the datasets.

Datasets for assessing the impact of class imbalances on repertoire label prediction and recovery of immune signals

Class imbalance in a binary classification problem occurs when one class significantly outnumbers the other. A total of 6 synthetic datasets are included for assessing the impact of class imbalances on repertoire label prediction and recovery of immune signals (**Figure 3.b**). Specifically, we included two different variations of class imbalance, where in one of the variations, the percentage of positive examples in training data was 20%, and in another variation 80%. Three replicated datasets were constructed for each of the variations. Using `simAIRR`⁶⁶, we generated a total of six datasets, where shared public sequences constitute the immune signals. In each dataset, known antigen-specific experimental sequences from VDJdb database⁹⁵ were used as immune state-associated features. For this, we used sequences specific to CMV. There was no overlap of immune signals between the datasets, once again, meaning that different sets of sequences were used as immune signals for each of the six datasets. A random sample of experimental sequences from ImmuneCode database⁷⁰ were used as the background sequences for repertoire construction. **Table S8** shows details of the characteristics that vary between the datasets.

Datasets for assessing the impact of systematic differences between comparator groups on repertoire label prediction and identification of immune signals

A total of 10 experimental datasets were assembled to assess the impact of systematic differences between comparator groups on repertoire label prediction and identification of immune signals (**Figure 3.b**). For this, we used three different experimental data sources: SARS-CoV-2-based cohort, T1D-based cohort and CMV-based cohort. Here, for each training dataset of 400 examples, a corresponding test set of 200 examples was used. There existed systematic differences between the data of different cohorts including library preparation differences. Specifically, the CMV-based cohort²⁵ used the Adaptive Biotechnologies immunoSEQ assay version hsTCRB v2 and v3, whereas the T1D-based cohort⁶⁹ used v4 and v4b. The SARS-CoV-2-based cohort⁷⁰ used v4. We assembled the 10 datasets while keeping either T1D+ or CMV+ as the positive class (immune state of interest for prediction), and varying negative examples either from the same study or from different studies. This resulted in datasets with or without systematic differences between positive and negative class examples. In this connection, we aimed to explore how the choice affects the ability to learn immune signals of interest. For this, we included different versions of test datasets for the same training dataset, where systematic differences between positive and negative class examples may or may not exist in the test datasets. A random sample of the desired number of examples from the original cohort was used to construct the datasets. See supplementary **Table S9** for details on how the datasets were assembled.

Negative control datasets

We included two negative control datasets (**Table S10**), where one is experimental and one is synthetic. Datasets that had previously been used for testing other attributes in Phase-2 have been selected to be modified as negative controls (**Figure 3.b**). We selected the datasets based on the performance of a baseline ML method (from pilot data shown later), where the baseline ML method already performed well on both the selected datasets. To serve the purpose of negative controls, we shuffled the labels of examples in both datasets.

Datasets for assessing the adaptability of top methods to BCR data

The architectures of some of the AIRR-ML methods (e.g., DeepRC⁹⁷), and even simpler approaches that rely on identification of convergent clonotypes can potentially be applied to both TCRs and BCRs (are often not stated as being restricted to only one of them, even though they are often assessed on only one or the other). Methods that rely on using protein language models (like ESM-2, Prot-5) for obtaining embeddings can also be readily applied to both TCRs and BCRs. It will thus be interesting to report on this aspect in the Phase 2, where we will internally evaluate all the top 10 methods on selected BCR datasets (particularly data from Zaslavsky et al., 2025³² or similar data that may become available at the time of stage 2 submission) without further customisation to comment on the relevance of shortlisted methods/approaches to immune receptors in general. It is interesting to investigate to what degree existing methodologies perform very differently on TCR vs BCR.

Data hosting and submission of predictions in Phase-2

For Phase-2 of the internal stratified performance analysis, the total size of the benchmarking database will approximately be 200 Gigabytes. The database will be shared with the method developers (those who want to train the models on their local infrastructure) as a download link through FileSender⁹⁸. The method developers will be asked to return a single compressed archive (e.g., zip file) containing predictions and

recovered immune signals of the datasets in the database in a pre-specified format, where templates of the pre-specified format will be provided upfront. As mentioned above, the developers will be allowed to return their submissions on a subset of the datasets if desired (on only one of the replications of each setting). In such a case, the variability of performance of such methods will not be presented in the detailed analysis.

Performance metrics in the Phase-2

We intend to use ROC AUC and Jaccard similarity as the main metrics for the quantification of classification performance and recovery of immune signals, respectively, in Phase-2. Note that these performance metrics are also used as the main metrics in Phase-1. In addition, other relevant metrics such as the balanced accuracy for repertoire label prediction, and the AUC for precision-recall curve for the recovery of immune signals will also be computed, analyzed, and reported in Phase-2. An overall ranking of the methods as well as ranking in individual categories will be reported.

Assessment of statistical significance of ranking of methods in the Phase-2

To assess the statistical significance of the ranking of methods in Phase-2, we will first perform a non-parametric Friedman test under the null hypothesis that the performance of all the ML methods are the same across all tasks and datasets. If the null hypothesis is rejected, we will perform post-hoc tests in a pairwise fashion to find out which methods have significantly different performances.

Leaderboards in the Phase-2

In Phase-2, we are not limited by platform constraints that require us to have a single leaderboard. Thus, in Phase-2, we will analyze, categorize and report results separately for Task 1 and Task 2.

Blinding the dataset characteristics in both phases

While this registered report transparently describes the variety of challenges included in the competition, the participants will be blinded to the specific assessment attributes to which each dataset is tied. We expect that such blinding reflects the real-world experimental datasets, as so far, prior knowledge about the immune signal characteristics from experimental datasets is limited.

Communication channel for the challenge participants in both phases

During Phase-1 of the Kaggle competition, the challenge participants will be able to discuss with other participants and communicate with the competition hosts through the discussion forum provided by the Kaggle platform itself. In Phase-2 of the internal stratified performance analysis, we will provide the method developers (who will also be co-authors) with a communication channel on Slack⁹⁹.

Process for assessing data suitability and establishing reference benchmarks

Conducting a pilot data analysis on the benchmarking database is crucial for several reasons outlined in the introduction. Particularly the suitability of the datasets to be included in the challenge has to be assessed. In addition, establishing reference benchmarks of performance on all datasets is valuable, as it provides

participants with a standard to compare against and improve upon. Therefore, we carried out a pilot data analysis on all the datasets included in the Kaggle competition (as described in **Tables 1 and 2**) and the later stage of stratified performance analysis (as described in **Tables S2-S8**). For this, we evaluated the performance on repertoire label prediction on all the datasets using a baseline method and a widely-used AIRR-ML method²⁵. In addition, we used two elementary approaches (one approach for each ML method) to quantify the level of interpretability (recovery of true immune signals) achievable by the ML methods on all the synthetic training datasets. In the sub-sections below, we first describe the details of these ML methods and interpretability approaches. Later, we describe the criteria and process for assessing data suitability and inclusion of benchmarking datasets in the database.

ML methods and interpretability approaches used for assessing the data suitability and establishing reference benchmarks of performance

For assessing the suitability of datasets for benchmarking and for establishing initial reference benchmarks of performance, we used one baseline ML method (logistic regression (LR)) and one widely-used AIRR-ML method²⁵ (hereafter referred to as the Emerson method). We used one elementary interpretability approach corresponding to each of the ML methods tested. Below, we describe the ML methods and interpretability approaches.

Logistic regression and the corresponding interpretability approach

As a baseline method, we chose to use a regularized LR approach, which has shown promise for certain types of immune signal patterns as described at length elsewhere⁶⁸. Briefly, for feature encoding, the amino acid sequences in each repertoire were split into overlapping subsequences of size=4 (4-mers), and their frequencies were calculated, followed by L2-normalization. Each feature vector was then standardized by scaling the variance to one and the mean to zero. We then trained scikit-learn's implementation of L1-regularized LR model on 4-mer encoded representation of amino acid sequences through immuneML version 2.1.2⁵⁵ for a binary classification problem of repertoire label prediction. The hyperparameters of the model (specifically regularization constant) were chosen through cross-validation. The assessment of label prediction performance was made on the supplied test data. For returning the top 50,000 sequences that likely harbor true immune signals, we used the coefficients of the trained LR model to score each unique sequence in the training data.

Emerson method and corresponding interpretability approach

As a representative example of a widely-used AIRR-ML method, we chose to use the approach of Emerson et al.²⁵ The Emerson method assumes full sequence identity in combination with V and J genes and assumes that immune state-associated sequences are observed with increased frequency (thus shared) in individuals that share the immune state (also referred to as public responses). Using Fisher's exact test, the method first identifies condition-associated sequences that are enriched in positive-labeled repertoires relative to negative-labeled repertoires. A generative binary classifier is then trained to infer the immune state from the number of condition-associated sequences within each repertoire. This two-step approach is akin to the MIL approach⁴⁵ introduced above, where the immune state-associated patterns are first determined followed by predicting a repertoire-level label based on some form of pooling function of the immune state-associated patterns. We trained the Emerson method using immuneML version 2.2.6⁵⁵. The hyperparameters of the model (specifically, the p-value threshold) were chosen through cross-validation.

The assessment of label prediction performance was made on the supplied test data. For returning the top 50,000 sequences that likely harbor true immune signals, we used the p-values of Fisher's exact test used by the Emerson method to rank the sequences and returned the top 50,000 sequences.

Criteria and process for assessing data suitability and dataset inclusion

To ensure the inclusion of datasets that represent an appropriate level of challenge for the ML methods, we assessed the suitability of datasets before including them in the benchmarking database. Here, we describe the criteria and process we followed for assessing data suitability and dataset inclusion. We used the initial performance benchmarks (**Figure 4, Figure 5, Figure 6, Figure 7**) as a guide to choose and include datasets that represent suitable challenges for the benchmarking. The criteria that we chose for being a suitable challenge are: we included the experimental datasets as they were, but chose to also include datasets that may potentially pose different challenges to the methods (autoimmune versus common infection as the primary immune state of interest for the ML methods). For synthetic datasets, instead of determining the suitability of each individual dataset in isolation, we determined the suitability of the collection of datasets specific to each assessment attribute as a whole. To ensure that the witness rates of synthetic datasets are "realistic" and sufficiently similar to those observed in real-world experimental data, we compared the performance of one baseline and one widely-used ML method on real-world data with their performance on synthetic data. Similar performance indicates a comparable level of problem difficulty. Our goal was not to create a single parameter configuration mimicking real-world data. Instead, while ensuring at least one configuration was realistic, we generated synthetic data at three difficulty levels: easy, moderate, and difficult (see **Figure 3.b**). By doing so, we could ensure that even baseline models worked well for the easiest versions, while they performed close to random on the most challenging versions. Including difficult datasets is crucial to push the boundaries of state-of-the-art methods through the community challenge. This three-level approach also facilitates post-hoc analysis of method behaviours. The criterion for synthetic datasets of each assessment attribute to represent a suitable challenge was that at least one of the tested methods should exhibit a moderate level of performance either on the repertoire label prediction or the recovery of true immune signals. We simulated the synthetic datasets of each assessment attribute in a feedback loop where, after simulation and initial performance benchmark, if any of the datasets in the category deemed less suitable, we discarded and repeated the simulation using different simulation parameters. We repeated this process as many times as required until a suitable dataset was simulated. Even if ROC AUC for repertoire label prediction was near random, we included datasets where methods began to pick up signals, as these represent suitable challenges with significant room for improvement for sophisticated methods. Pilot data (Figures 4-6) indicate that the chosen ML methods do not exhibit good performance on a large majority of the synthetic datasets, suggesting the included datasets are not biased in favour of the tested ML methods. Notably, Emerson method, conceptually similar to those in FDA-approved AIRR-based diagnostic tests, performs poorly on most synthetic datasets. This poor performance highlights a significant margin for improvement, underscoring the innovation we aim to foster through this community challenge.

Although one of the tested approaches exhibited superior performance on some of the included datasets, we still chose to include different datasets because not all operating assumptions and architectural choices of methods are necessarily equally suited to diverse types of immune signal patterns and study design characteristics. As a consequence, a method may not perform equally well on all the datasets in the benchmarking database. Therefore, to analyze which operating assumptions and architectural choices of

methods are best suited to different challenges of repertoire label prediction and recovery of immune signals, it is essential to include datasets of varying difficulty in the database.

Pilot data

Assessing data suitability and establishing reference benchmarks

To reiterate the need and significance of the pilot data within the context of this study, we revisit key points detailed above in the introduction. The primary objectives of pilot data analysis include: (a) Assessing dataset suitability, (b) Ensuring varying degrees of difficulty, (c) Providing Reference Benchmarks of Performance, and (d) Evaluating the value of sophisticated methods. To address these needs, we generated pilot data to evaluate the performance of both immune state prediction and recovery of immune state-associated receptors across all datasets in the benchmarking database using a baseline method and a widely used AIRR-ML method. Each sub-section below describes the rationale for including the respective assessment attributes in the challenge and provides pilot data on reference benchmarks of performance as well as data suitability.

Datasets assessing the impact of signal abundance and diverse signal patterns on repertoire label prediction and recovery of immune signals

One of the fundamental characteristics of adaptive immune responses in different immune states (e.g., common infections, autoimmunity, cancer) is how abundant and detectable the condition-associated signals are, as encoded in the repertoires in various observed or latent forms. For instance, in common infections like CMV, the accumulation of large numbers of antigen-specific CD4⁺ T cells is known and often reported¹⁰⁰, whereas the immune signals in other conditions like autoimmunity and cancer were initially found to be modest^{38,79,80}. Moreover, the immune signal information encoded in the receptor sequences has been observed in distinct forms: public sequences were most helpful for discriminating some immune states^{25–27}, while enriched short sub-sequence patterns (motifs) were more helpful in discriminating others^{34,84–87}. Therefore, as part of the benchmarking competition, it is crucial to investigate the signal abundances required to build robust diagnostic models and maximize the discovery of immune state-associated features, and to determine if these requirements vary based on the different signal patterns. For this, we included experimental and synthetic datasets that will represent diverse signal patterns and different levels of signal abundance in both the Kaggle competition and the subsequent stratified analysis phase.

Reference benchmarks of performance: We noticed that the LR approach performed well on the high witness rate synthetic datasets where the immune signal was constituted by k-mers (e.g., ROC AUC of 1.0 at 0.4% witness rate) (**Figure 4.a** and **4.b**), and performed moderately well on the experimental **T1D** test sets (ROC AUC 0.69 ~ 0.76). LR approach performed poorly on the CMV and HSV-2 test sets, where the ROC AUC was 0.59 and 0.47 respectively (**Figure 4.i** and **4.j**). The Emerson method²⁵ performed well on the high witness rate synthetic datasets where the immune signal consisted of public sequences (e.g., ROC AUC of 1.0 at 0.4% witness rate) (**Figure 4.e** and **4.f**), and also performed moderately well on the experimental test sets (ROC AUC 0.61 ~ 0.84) (**Figure 4.i** and **4.j**). As expected, the LR approach performed poorly on synthetic datasets where the immune signal consisted of public sequences, and the

Emerson method performed poorly on synthetic datasets where the immune signal consisted of k-mers both in repertoire label prediction and interpretability. While both the methods performed modestly in recovering the true immune signals even at the highest witness rate of 0.4% (**Figure 4.c, 4.d, 4.g. and 4.h**), the interpretability attainable on synthetic datasets, where the immune signal was comprised of k-mers, was noticeably lower (**Figure 4.c and 4.d**). However, we observed a very high correlation between interpretability and classification performance for both methods (Pearson correlation coefficient > 0.97). We included a negative control experiment⁹⁴ for both the synthetic data and the experimental data (one each from **Figure 4.f and 4.i**, respectively) to confirm that the observed good performance of the tested methods on datasets in **Figure 4** is indeed due to learning the immune signals associated with the repertoire labels. For this, we shuffled the labels of an experimental T1D dataset on which both of the tested ML methods exhibited a moderate level of performance (ROC AUC 0.78 and 0.79 respectively), and a synthetic dataset of high witness rate (0.4%), where almost perfect classification was found with the Emerson method shown in **Figure 4.f**. As expected, performance of both of the tested ML methods dropped from near-perfect classification to close to random in the negative control data (**Figure 4.k**). Furthermore, while the Jaccard similarity on the original synthetic dataset with a witness rate of 0.4% was approximately 0.58, it dropped to 0.009 in the negative control dataset with shuffled labels.

Data suitability: The synthetic datasets represented diverse signal patterns, where the pilot data demonstrated that only methods with matching operating assumptions performed well. The differences in performance of the method that relied on k-mer encoding on experimental datasets representing different immune states (common infection, autoimmune) also indicated that the different immune states may harbor different signal patterns. This, in the community challenge, will aid the investigation of whether the state-of-the-art methods are able to encapsulate the full spectrum of diverse signal patterns and thus, perform equally well for diverse signal patterns and immune states. The pilot data further demonstrated that the datasets represented varying degrees of difficulty, where only the methods with matching operating assumptions of true immune signals performed well in the easiest setting of high witness rate while performing moderately or close to random at low witness rates. The varying degrees of difficulty was also visible in the recovery of true immune signals. In the community challenge, this will allow the analysis of which methodological choices and approaches of particular methods aid in learning equally well on low signal abundances. Interestingly, none of the interpretability approaches for either of the methods performed well, and the community challenge is thus seen as a significant opportunity to discover approaches that maximize the recovery of true immune signals. In addition, the availability of covariate information including HLA, age, sex and library preparation methods in T1D cohort allows us to retrospectively investigate (after the conclusion of the challenge) whether the top methods generalize well within the disease population stratified by different confounders. Overall, the pilot data demonstrated that the datasets represented suitable challenges with diverse signal patterns and varying degrees of difficulty in terms of signal abundances.

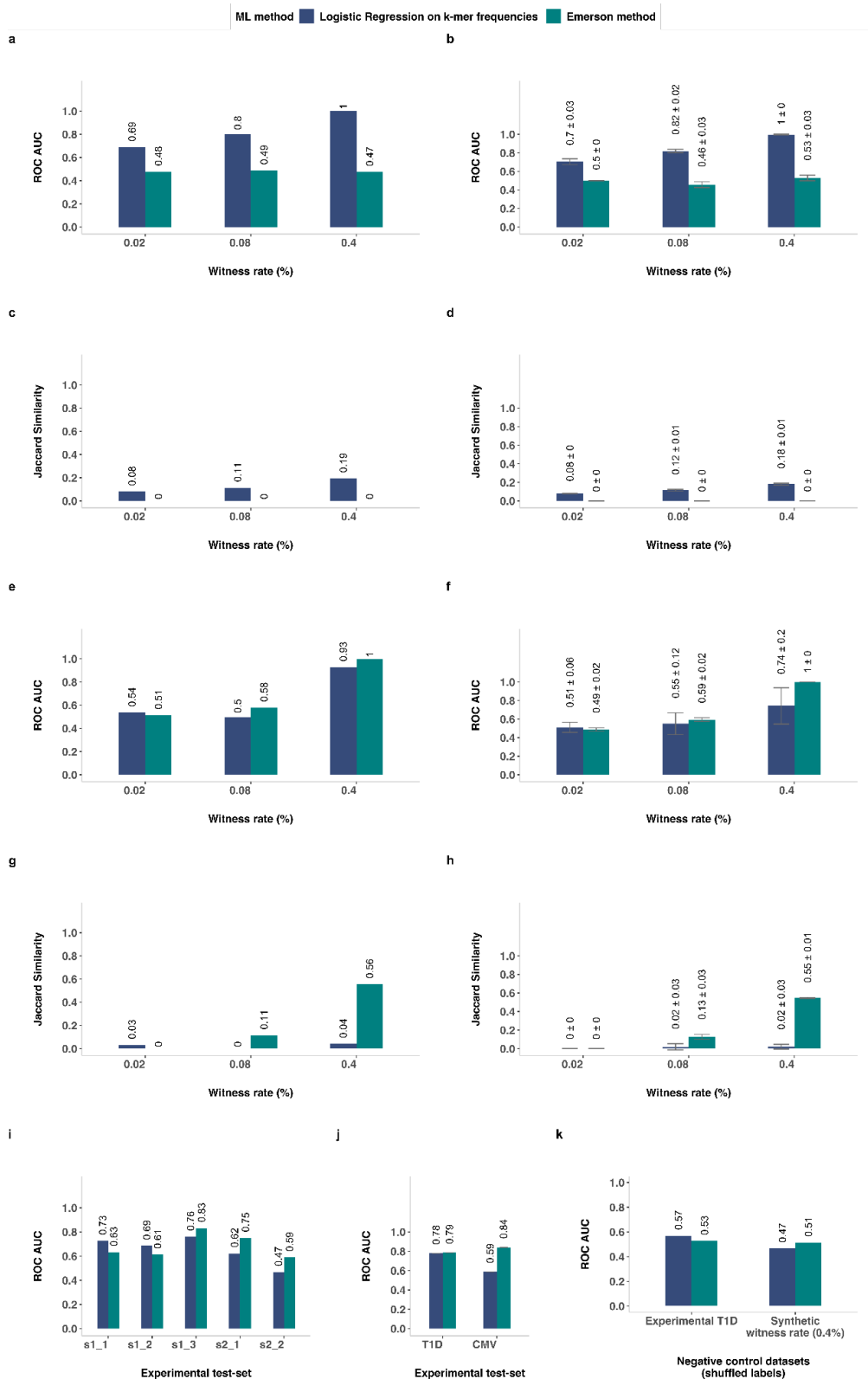


Figure 4 Pilot data on the evaluation of datasets assessing the impact of signal abundance and diverse immune signal patterns using two ML methods. Performance of two different ML methods (LR on k-mer frequencies and Emerson method²⁵) are shown in two different colors. In all the sub-panels (a-k), the y-axis represents the performance measures for either repertoire

label prediction (ROC AUC) or interpretability (Jaccard index) and the x-axis represents either the witness rate for synthetic datasets (a-h) or different test sets for experimental datasets (i, j) and negative controls (k). Note that the experimental datasets representing distinct immune states (infection vs autoimmune condition) are presumed to represent different challenges for the ML methods in terms of witness rate (j). The reference benchmarks of performance of two ML methods are shown on synthetic datasets, where enriched sub-sequence patterns (a-d) or public sequences (e-h) constitute the true immune signals and on experimental datasets representing distinct immune states (i, j). The findings on datasets included in the Kaggle competition (a, c, e, g and i), and those included in Phase-2 of stratified performance analysis (b, d, f, h, j and k) are shown separately. The error bars indicate standard deviation of the performance as calibrated on three independent replications of the analysis. Note that the datasets included in Kaggle competition and panels j, k do not show error bars because of using a single replication. **Training and test dataset sizes:** 400 examples each in training and test with balanced labels were included in all synthetic datasets (a-h) and for negative controls (k). For experimental datasets in the Kaggle competition (i), 900 examples for s1 training with ~35% positive label rate and ~400, 400, and 800 examples each in test sets s1_1, s1_2, and s1_3 respectively with varying class balances (not revealed to avoid information leakage) were included. For s2, 306 examples were included for training and 76, 100 examples were included for s2_1 and s2_2 test sets. For experimental datasets of phase-2 (j), 400 examples each were included in training and test with balanced labels in all the datasets except CMV dataset, where only 200 examples were used in the test split.

Datasets assessing the impact of the size of training data and species richness on repertoire label prediction and recovery of immune signals

From a study design perspective, the ability of discriminative models in detecting and learning immune signals depends on the size of available training data (number of individuals tested; subjects) and the species richness resulting from sampling depth^{15,16,44,101}. Note that species richness here refers to the number of unique clonotypes observed in a sequenced repertoire. A key challenge is that a very small fraction of AIRR studies in general comprise more than 100 subjects, and often the species richness is low. For instance, as of October 02, 2024, <5% studies curated in the AIRR Data Commons^{91,92} have > 100 subjects, and among the studies that have a minimum of 50 subjects, > 75% of studies have an average species richness < 70,000. Previous studies showed that repertoire classification methods generalize poorly from smaller training data sizes^{25,55,87}. Therefore, there is a need to push the boundaries of the state-of-the-art to generalize well from smaller training data sizes. In addition, the scientific community would benefit by knowing what sampling depth, and resulting species richness, is sufficient and required for developing generalizable diagnostic models and discovering antigen-specific features. To answer these questions, we included experimental and synthetic datasets that will represent different sizes of training data and species richness.

Reference benchmarks of performance for training data size: We noticed that the recovery of immune signals was generally poor on the synthetic datasets, irrespective of the training data size, although there was an increase in the recovery with an increase in the dataset size (**Figure 5.d**). We noticed that both of the tested ML methods performed poorly in repertoire label prediction on training dataset sizes < 400 in both experimental CMV and synthetic datasets (**Figure 5.b** and **5.c**), whereas they performed moderately well on the experimental T1D datasets irrespective of the training dataset size (**Figure 5.a**). Notably, a training dataset size that is more than twice the size used in this pilot analysis (~ 900 examples in the Kaggle competition versus 400 examples in this analysis) did not yield remarkably superior separation of the T1D examples with the same ML methods (**Figure 4.i**; referring to the ROC AUC of 0.76 and 0.83, respectively, by LR and Emerson method on experimental test set 3 in the Kaggle competition). Although the performance of Emerson method at a training dataset size of 400 is relatively modest in the synthetic datasets (when compared to the same setting in experimental datasets), the moderate level of recovery of immune signals at this setting hints at the potential of sophisticated approaches that may generalize well

from smaller training dataset sizes. We observed a Pearson correlation coefficient of 0.57 between interpretability and classification performance for the Emerson method on synthetic datasets.

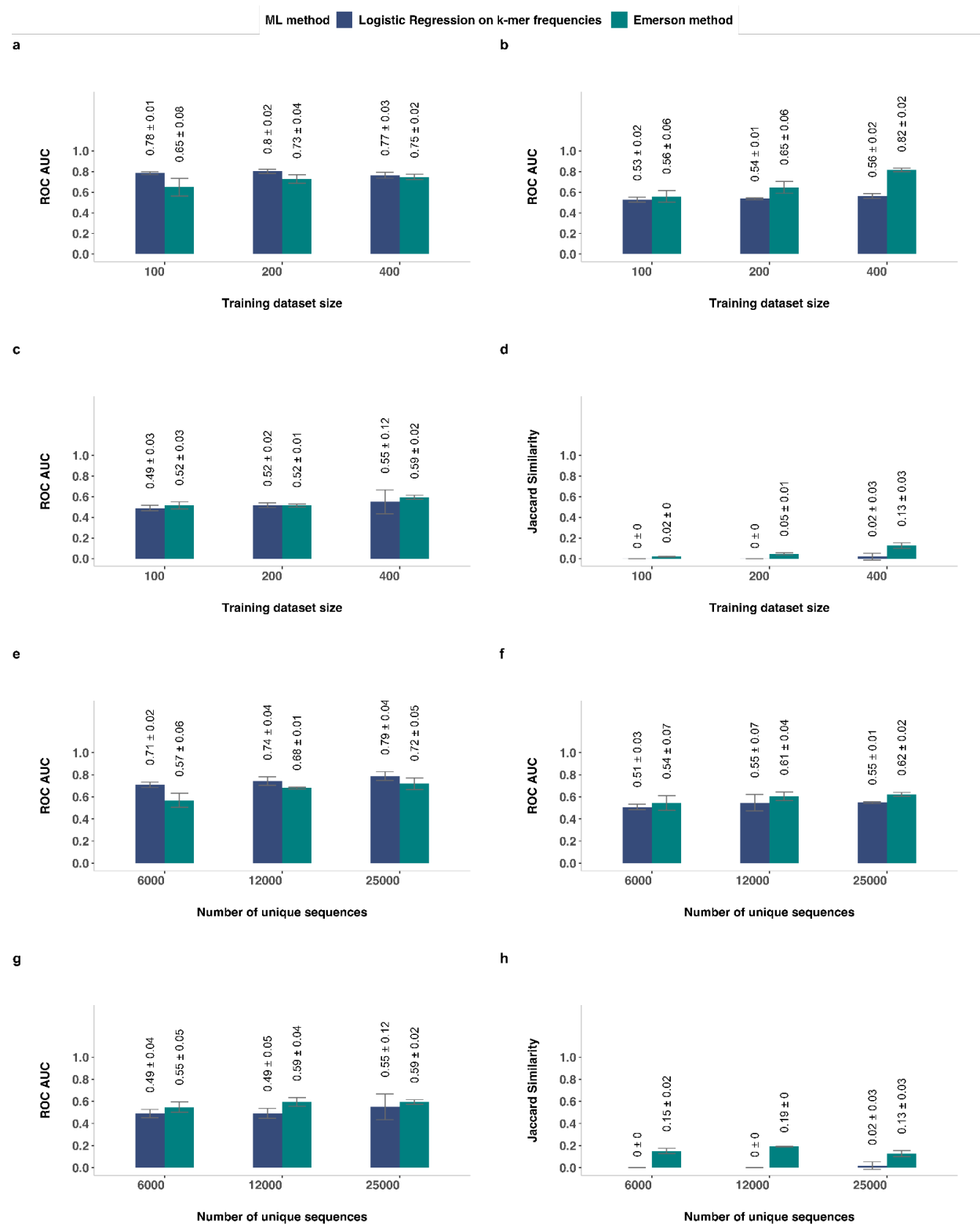


Figure 5 Pilot data on the evaluation of datasets assessing the impact of the training dataset sizes and species richness using two ML methods. Performance of two different ML methods (LR on k-mer frequencies and Emerson method²⁵) are shown in two different colors. In all the sub-panels (a-h), the y-axis represents the performance measures for either repertoire label prediction (ROC AUC) or interpretability (Jaccard index) and the x-axis represents either the training dataset size (a-d) or the number of unique sequences within each training repertoire (species richness) (e-h). The reference benchmarks of performance of two ML methods are shown for experimental T1D-based datasets (a, e), experimental CMV-based datasets (b, f), and synthetic datasets with public sequences constituting the true immune signals (c, d, g, h). The error bars indicate standard deviation of the performance as calibrated on three independent replications of the analysis. The Signal abundance level was maintained at 0.08% across all the synthetic datasets. Both training and test splits had similar average species richness across all datasets. **Training and test dataset sizes:** 400 examples each in training and test with balanced labels were included in all datasets except when varying the training dataset sizes and for the test dataset of CMV-based datasets, where only 200 examples are included.

Reference benchmarks of performance for species richness: We noticed that on experimental T1D datasets, an unexpectedly high level of performance on repertoire label prediction was achieved even at species richness as low as 6000 unique sequences with LR on k-mer frequencies (Figure 5.e). Although the performance of both the ML methods increased with an increase in species richness on T1D datasets, it was particularly noticeable that LR on k-mer frequencies performed better. With experimental CMV datasets and synthetic datasets, such a level of performance was not generally visible across the analysed settings (Figure 5.f and 5.g). Here, it is noteworthy that on an experimental CMV dataset with similar training dataset size but with an average species richness of 200,000 instead of the 25,000 sequences analysed here, an ROC AUC of 0.84 was observed with Emerson method (Figure 4.j). A moderate level of recovery of true immune signals was observed on the synthetic datasets, alluding to the potential of innovative approaches faring better at both repertoire label prediction and recovery of immune signals. We observed a Pearson correlation coefficient of 0.40 between interpretability and classification performance for the Emerson method on synthetic datasets.

Data suitability: The performance of both the tested methods differed clearly between distinct immune states (T1D and CMV), where the performance was moderate on T1D datasets while being poor on CMV datasets across the analysed settings (Figure 5.a vs 5.b and 5.e vs 5.f). The performances of both the tested methods on synthetic datasets were also modest. Despite this, we observed that the performance of Emerson method²⁵ increased with an increase in either training dataset size or species richness across different types of datasets, both for repertoire label prediction and recovery of true immune signals. All these observations together indicate that the datasets represent varying degrees of difficulty for the ML methods. In the community challenge, this will allow the analysis of which methodological choices and approaches of particular methods aid in generalizing sufficiently well from smaller training data sizes and lower species richness. The findings of the competition can also reveal whether an increase in training dataset sizes and species richness beyond the parameters tested might have additional benefits. The performance of both the tested methods was overall modest across settings, indicating considerable potential for improvement in repertoire label prediction and the recovery of true immune signals. Overall, the pilot data demonstrated that the datasets represented suitable challenges with varying degrees of difficulty in terms of varying training data sizes and species richness, as well as differences based on distinct immune states.

Datasets assessing the impact of signal abundance shifts and class imbalances on repertoire label prediction and recovery of immune signals

Among other common study design-related aspects that can affect ML systems in a variety of ways are dataset shifts^{102–104} and class imbalances^{105–107}. The target domains can often change in a variety of ways

from the source domain in real-world applications (particularly in medicine), leading to deterioration in performance of the ML methods. In the context of AIRR data, source domain and target domain can systematically differ in many aspects such as time since exposure, regions, genetic ancestry, sampling issues (convenience sampling, selection biases), protocol differences, etc.¹⁰⁸. Failing to generalize to out-of-distribution data is often attributed to not learning the causal factors of variation in the data and instead learning spurious correlations in the source domain¹⁰⁸. Since none of the existing AIRR-ML methods in the literature were devised to readily adapt to out-of-distribution data³⁸, there exists a critical gap of knowledge specific to the AIRR domain. We therefore included datasets to investigate if ML methods are able to learn the causal factors at different signal abundance levels, and how do they generalize to a target domain when the signal abundance drops.

Another major study design aspect that needs a recommendation based on benchmarking the behavior of ML methods is the class balance. Class imbalance and the resulting underrepresentation of a class is a known study design issue, particularly in a field like medicine (e.g., because of less prevalence of a disease in a population, less responses from healthy volunteers to give samples etc.). Although, the impact of class imbalance on ML methods is well-known^{105,106}, knowledge on the following aspects is extremely useful for the AIRR study design: (a) what level of disproportion (in relation to the positive class of interest) can the ML methods tolerate and generalize well for repertoire label prediction, (b) is a particular way of disproportion (positive class majority versus positive class minority) better for repertoire label prediction and recovery of immune signals, (c) what approaches to handle imbalances can the field adopt as a standard.

Reference benchmarks of performance for signal abundance shifts: We included three settings: (a) one setting intended for comparison where there was no shift in signal abundance between training and test and maintained at 0.08%, (b) one where signal abundance in training was 0.16% and dropped to 0.08%, and (c) one where signal abundance in training was 0.08% and dropped to 0.02%. We noticed that the separation ability of the Emerson method was good when the signal abundance shifted from 0.16% to 0.08%, but dropped when the signal abundance shifted from 0.08% in training to 0.02% in test data (**Figure 6.a**). Notably, the performance was similar even when there was no signal abundance shift (constant at 0.08%) and when there was a shift (from 0.08% to 0.02%), albeit with a larger variance. The uncertainty of ROC AUC was large when signal abundance shifted from 0.08% to 0.02%. Interestingly, unlike when there was no signal abundance shift, none of the examples in the test set were predicted as positive examples in both the cases of signal abundance shift; thus, the recall of the model was zero (**Figure 6.b**) and the precision was undefined (**Figure 6.c**). As expected, the performance on the recovery of true immune signals dropped with decrease in signal abundance in the training data. Overall, we observed a Pearson correlation coefficient of 0.58 between interpretability and classification performance for the Emerson method on synthetic datasets.

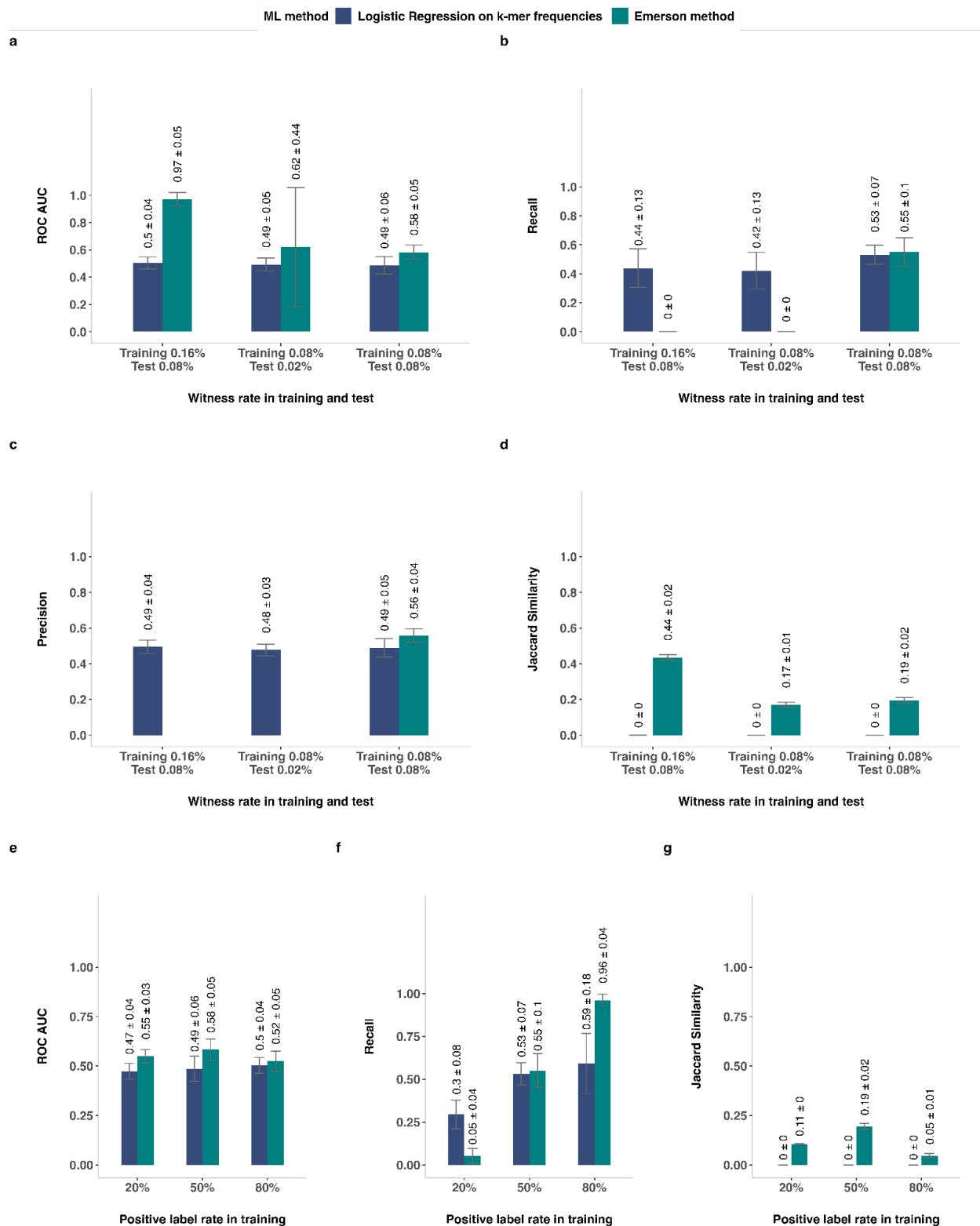


Figure 6 Pilot data on the evaluation of datasets assessing the impact of signal abundance shifts and class imbalances on repertoire label prediction and recovery of immune signals. Performance of two different ML methods (LR on k-mer frequencies and Emerson method ²⁵) are shown in two different colors. In all the sub-panels (a-g), the y-axis represents the performance measures for either repertoire label prediction (ROC AUC, recall, precision) or interpretability (Jaccard index) and

the x-axis represents either the witness rates in training and test datasets (signal abundance shifts) (a-d) or the positive label rate indicating class balance (e-g). The error bars indicate standard deviation of the performance as calibrated on three independent replications of the analysis. A constant witness rate of 0.08% is used across the datasets assessing the impact of class imbalance. **Training and test dataset sizes:** 400 examples each in training and test were used across all the settings analysed.

Reference benchmarks of performance for class imbalances: We included three settings: (a) one with balanced labels for comparison, (b) one where the positive label rate was 20% (1:5), and (c) one where the positive label rate was 80% (4:5). The signal abundance level was maintained at 0.08% across all the settings. While the separation ability of the models was generally low at this chosen moderate witness rate of 0.08%, the performance of both repertoire label prediction and recovery of immune signals was highest at the setting of balanced labels (Figure 6.e). When the positive label rate was 20%, the Emerson method predicted the majority of examples in the test data as negative and vice versa. When the positive label rate was 80% instead in the training data, that reflected through the recall of the model on test data (Figure 6.f). Interestingly, the interpretability measure, though low overall, was twice as good when the positive class was in the minority compared to when it was in the majority (Figure 6.g). Overall, we observed a Pearson correlation coefficient of 0.48 between interpretability and classification performance for the Emerson method on synthetic datasets.

Data suitability: The well-performing model differed in performance between optimal settings for the model (training on high witness rate and balanced labels) and the comparison settings (Figure 6.a-g). Moreover, the behavior of the well-performing model differed when there was a signal abundance shift versus no signal abundance shift as evident through precision and recall on the test datasets (Figure 6.b and 6.c). Similarly, the recall and interpretability of the well-performing model differed when the positive class was the majority or minority in the training data (Figure 6.f and 6.g). Taken together, these observations indicate that the datasets represent varying degrees of difficulty for the ML methods. In the community challenge, this will allow the analysis of which methodological choices aid in adapting to signal abundance shifts and generalizing equally well across all the settings analyzed, including various forms of class imbalances. In addition, the analysis may provide useful recommendations on study design aspects to the field. The performance of both the tested methods was overall modest across settings, indicating considerable potential for improvement in repertoire label prediction and the recovery of true immune signals. Overall, the pilot data demonstrated that the datasets represented suitable challenges with varying degrees of difficulty in terms of signal abundance shifts and class imbalances.

Datasets assessing the impact of systematic differences between positive and negative class examples

Often in biology and medicine, controls (comparison group) are reused from a different study (extracted from public databases) owing to experimental costs, lack of sufficient responses from control groups, ethical considerations, etc. When systematic differences exist between positive and negative examples, ML methods may learn the systematic differences instead of or in addition to the true signals of interest. In this regard, we designed ten different training datasets representing two primary immune states of interest (positive class): T1D+ and CMV+, where the negative class examples either come from the same study with minimal systematic differences or from a different study with clear systematic differences (e.g., sequencing protocols). Different ML methods can exhibit different behaviors on these datasets depending on specific assumptions, inductive biases, feature representations, etc. The objectives of addressing this

question were to generate knowledge for the field on whether systematic differences between positive and negative examples are tolerated by the ML methods, whether the ML methods (depending on their specific choices) are able to learn the true signals instead of the systematic differences, and if they do, which specific choices enabled the increase in signal-to-noise detection ratio.

Reference benchmarks of performance: We noticed that both the tested ML methods exhibited largely similar behavior except for minor differences in performances on individual datasets. The most striking observation was that the methods exhibited almost perfect separation ability when the training and test datasets had similar systematic differences between positive and negative examples, than when the test datasets did not have systematic differences between protocols. This observation was consistent across the primary immune states of interest, T1D and CMV (**Figure 7.a** and **7.b**). Specifically, when the training and test datasets both include either CMV+ vs. SARS-CoV-2+ or CMV+ vs. T1D-, the separation was almost perfect – whereas the performance dropped when the test datasets instead comprised CMV+ vs CMV-. Similar observations were made with T1D, where the models struggled to separate T1D+ and T1D- well, while excelling in separating T1D+ vs. CMV- and T1D+ vs. SARS-CoV-2+. When we checked the agreement between the models' top explanatory sequences, there was little overlap in general except for the models where CMV+ is the primary immune state of interest using Emerson method (**Figure 7.c-f**).

Data suitability: The constructed combinations of training and test datasets using distinct immune states with or without systematic differences between positive and negative class examples showed utility (based on pilot data) in demarcating whether or not the methods are able to learn the immune-state-associated signals of the primary immune state (positive label) in the presence of systematic differences. The observed differences in performances of the tested methods on the constructed datasets indicated that the datasets represented varying degrees of difficulty, where the training dataset containing examples with systematic differences but not the test dataset are considered the most difficult. In the community challenge, this will allow the analysis of which methodological choices and approaches of particular methods enable the learning of true signals over noise. In addition, it remains to be seen if any of the models in the community challenge can obtain larger agreement between models' interpretability (**Figure 7.c-f**) and what methodological choices enable that. Overall, the pilot data demonstrated that the datasets represented suitable challenges with varying degrees of difficulty in terms of systematic differences between positive and negative class examples.

Discussion

Neutral benchmarking studies, which avoid the "self-assessment trap", are crucial for advancing computational methods in many scientific disciplines^{109–112}. The established and well-described core principles of benchmarking^{51,113,114} emphasize that selecting appropriate datasets is a critical design choice. There is a consensus that a combination of real-world and simulated data represents a comprehensive design choice for benchmarking studies due to their respective strengths and limitations^{51,113,114}. As we have previously argued, synthetic data should almost always complement real-world experimental data⁴⁹. Relying solely on real-world experimental data has several drawbacks: (a) It carries a high risk of drawing unwarranted conclusions about uncertainty estimates from datasets of limited size. (b) It is impractical to thoroughly test methods' sensitivity to variations in problem configurations at scale, including robustness to domain shifts. (c) It is often impossible to discern whether patterns extracted by a method reflect

underlying causal relationships. (d) Spurious correlations can arise from suboptimal experimental designs (e.g., batch effects, temporal effects, selection bias, or demographic representation), which top-performing methods can at least partially exploit. (e) most importantly, it is often impractical to have access to ground truth. Synthetic data also has limitations: (a) It may not fully capture the complexity of real-world data. (b) It might lack the noise and biases present in real-world data. (c) Assumptions on ground truth signals may not necessarily align with real-world data unless sufficient established knowledge exists. While real-world experimental data inherently avoids the limitations specific to synthetic data, synthetic data almost always helps overcome the inherent limitations of real-world experimental data. Therefore, a comprehensive benchmarking approach requires the complementarity of real-world experimental data and synthetic data.

In this study, we ensured an equal representation of real-world experimental and synthetic data in our benchmarking database. For each assessment attribute, we included synthetic data to complement real data. This provides controlled settings with known ground truth, allowing us to determine if methods perform well by learning label-associated patterns. Findings from these additional controlled settings will only enhance the quality of our conclusions. Without synthetic data, it would be impractical to establish ground truth in diverse scenarios, making it impossible to evaluate method performance on Task 2 (recovering label-associated immune signals). Furthermore, without synthetic data, it's impossible to ascertain whether observed good performance stems from methods learning genuine signals or merely exploiting stable experimental artifacts in train/test splits. Thus, the current volume of synthetic datasets is essential to complement the experimental datasets and enable meaningful conclusions for each assessment attribute.

Acknowledging the limitations of synthetic data, we implemented several precautions to minimize their weaknesses: (a) We have invested substantial effort over the years in exploring, understanding, and developing simulation frameworks^{39,66} to ensure they faithfully represent the characteristics and statistical properties of experimental AIRR datasets with high fidelity, while avoiding the introduction of spurious artifacts. (b) Except for one deliberate setting, we used real-world experimental receptor sequences as a pool for randomly sampling background sequences when constructing repertoires. (c) We consciously avoided any systematic differences between positive and negative class examples during data generation. (d) Instead of relying on a single assumption for ground truth immune signals (which may or may not match real-world data), we used many variations of assumptions on immune signals to test if methods succeed in learning all such patterns. (e) In many settings, instead of hypothesizing on potential immune signals and enriched motifs, we added known antigen-associated sequences from experimental sources as ground truth. This avoids assumptions about their similarity, structure of patterns, or whether they contain enriched motifs. While we do not claim these precautions make the synthetic data perfectly match real-world biological complexity, they collectively minimize the weaknesses associated with synthetic data.

The large number of unique train/test sets in Phase 2 (~ 100) result from having three independent replications of the same setting in most cases to obtain good estimates of uncertainty of methods' performance. Without good quantification of uncertainty estimates, there is a risk of observing and reporting findings resulting from external factors such as parameter choices or over-tuning rather than from true biological relevance. If we ignore the replications and negative control experiments, the unique settings represent different parameter configurations (or difficulty levels) in six broad categories (signal abundance, different immune signal patterns, dataset sizes etc). All the comparisons will be within the six broad categories. Different parameter configurations are needed within the broad categories for comprehensive

benchmarking. In the community challenge, this will allow the analysis of which methodological choices and approaches of particular methods aid in learning well on low signal abundances and the full-spectrum of immune signal patterns, generalizing sufficiently well from smaller training data sizes and lower species richness, adapting well to signal abundance shifts and robust to various forms of class imbalances, enabling the learning of true signals over noise etc. We are confident that despite the initial impression of a large number of unique train/test sets, our structured approach, including replication settings and clear categorization within relevant assessment attributes, will indeed allow for the extraction of highly valuable and actionable insights.

Data availability

All the synthetic datasets described in this manuscript will be made available to the challenge participants either through the Kaggle platform (Phase-1) or through a downloadable URL from FileSender⁹⁸. The synthetic datasets will also be shared through AIRR Data Commons^{91,92} after the conclusion of the challenge. The published experimental data sources used in this study are available through public databases. The CMV cohort is available through Adaptive Biotechnologies ImmuneAccess platform at <https://doi.org/10.21417/B7001Z>. The SARS-CoV-2 cohort is available at <https://doi.org/10.21417/ADPT2020COVID>. Both of these datasets are also available through the AIRR Data Commons^{91,92}. The unpublished T1D cohort will also be made available through the AIRR Data Commons^{91,92} with a private embargo until the conclusion of the competition. All the datasets used in this study will be frozen with a permanent DOI at NIRD research data archive under open access license, but will remain private until the conclusion of the challenge.

Code availability

All the code used to generate the datasets and conduct pilot data analysis is hosted at <https://github.com/uio-bmi/AIRR-ML-BM-2025>. The repository will remain private until the conclusion of the challenge to not reveal the characteristics and labels of the datasets to the challenge participants. We will use the same repository to host all the code related to the analyses of Phase-1 and Phase-2. All code will be shared once the stage 2 manuscript has been accepted. To win the prize money in the Kaggle competition, the participants are required to share their methods openly. The methods that will participate in Phase-2 will be asked to share a containerized version of their model code, adhering to a uniform interface of running the models.

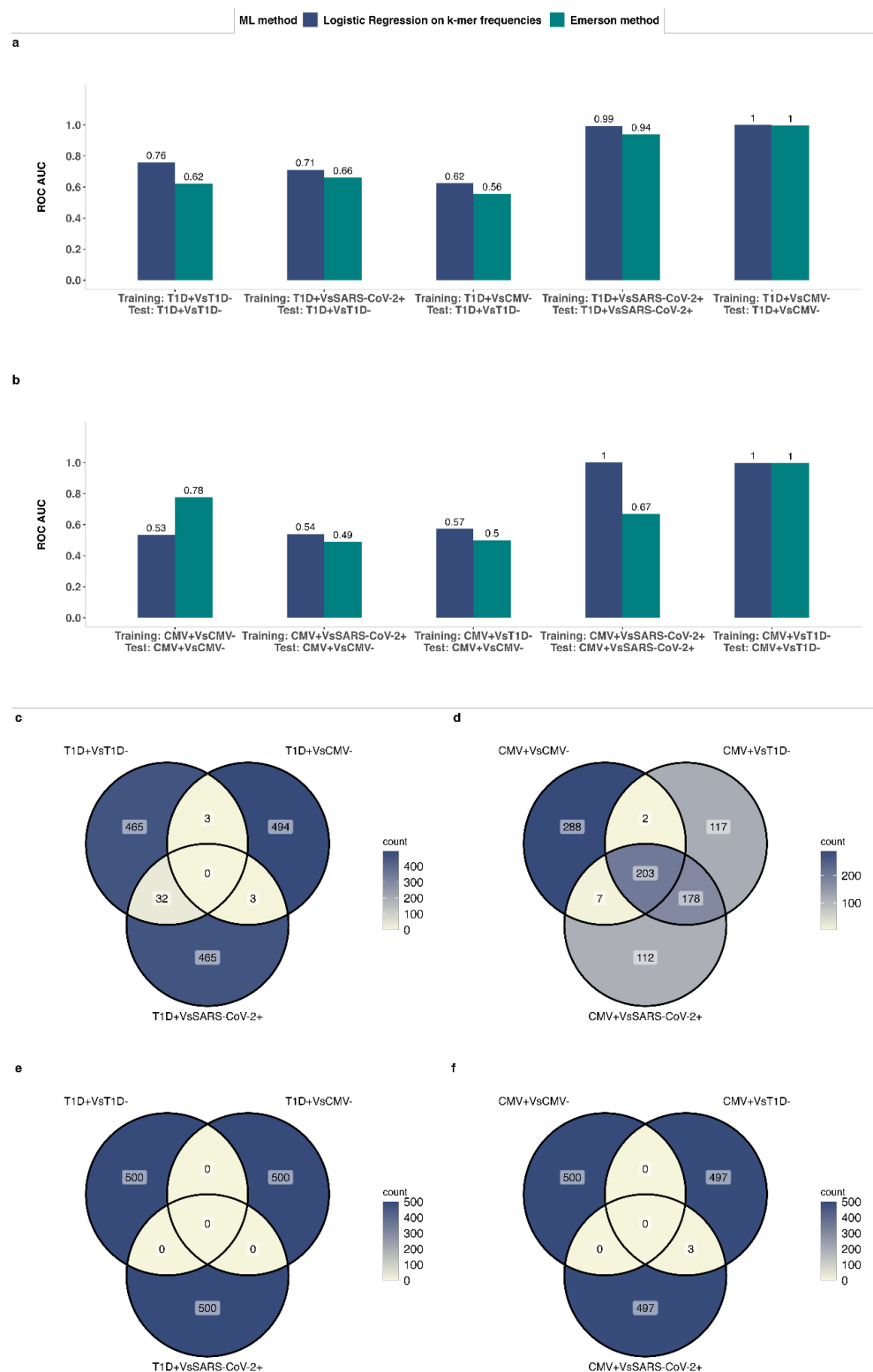


Figure 7 Pilot data on the evaluation of datasets assessing the impact of choosing control repertoires⁹³ from a different study (e.g. with different protocols and different immune states) on ML model performance. The performance on repertoire label prediction (ROC AUC on y-axis of **a**, **b**) of two different ML methods (LR on k-mer frequencies and Emerson method²⁵) is shown in different colors for different training and test data combinations (on x-axis of **a**, **b**), where the immune states of interest

1 (positive class) in the trained models varied (T1D in **a** ; CMV in **b**). The different training and test data combinations (indicated on
 2 x-axis of **a, b**) either differ or not differ in terms of protocols. The venn diagrams (**c-f**) show the overlap between the top explanatory
 3 features (top 500) of the different diagnostic models trained either using Emerson method (**c, d**) or the LR model (**e, f**). The colors
 4 in the venn diagrams (**c-f**) indicate the counts as shown in the count scales.

5 Glossary

Adaptive immune receptors (AIR)	T cell receptors (TCRs) and B cell receptors (BCRs) expressed by T lymphocytes and B lymphocytes, respectively ¹¹ .
Adaptive immune receptor repertoire (AIRR)	The collection of adaptive immune receptors (AIRs) sampled from a single individual at a fixed time point.
Architectures and architectural choices (of ML methods)	This refers to the fundamental design and structure of a ML model. Just as different types of buildings (e.g., a house vs. a skyscraper) have different architectures for different purposes, different ML models have different architectures (e.g., neural networks vs. decision trees) that make them suitable for different types of problems and data.
Assessment attributes	Specific characteristics or factors can influence how well a ML method performs. By varying these attributes, one can systematically evaluate the behavior and capabilities of different ML methods under various conditions. Examples from the AIRR field include label-associated sequence patterns, signal abundance, and study design characteristics (e.g., dataset size, species richness, (number of unique clonotypes), signal abundance shifts, class imbalances etc.
Background sequences (in the context of AIRR data)	Sequences that are not associated with a particular immune event of interest in AIRR datasets.
Balanced accuracy	This is a performance metric used in classification, especially when dealing with imbalanced datasets (where one class has significantly more examples than others). It's calculated as the average of specificity (true negative rate) and recall (true positive rate) for each class, providing a more fair assessment than simple accuracy when classes are unbalanced.
Baseline performance	Minimum level of acceptable performance generated independently for each dataset using simpler models intended to aid with comparison of relative improvements in performance of other models ¹¹⁵ .
Biases (in data)	In ML, "biases" refer to systematic errors or skewed distributions in the data that can lead a model to make unfair or inaccurate predictions. These can arise from how data is collected (e.g., differences in lab procedures) or from demographic imbalances within the dataset (e.g., disproportionate representation of certain age or sex groups).

Binary classification problem	This is a type of ML problem where the goal is to categorize examples into one of two possible outcomes or classes. For example, predicting whether a patient has a disease (positive) or not (negative) is a binary classification problem.
Biological interpretability	This term refers to how well we can understand the biological meaning or reasons behind the decisions or patterns identified by a ML model. For example, if a model predicts a disease, biological interpretability would mean being able to identify which specific biological factors (like certain immune receptor sequences) led to that prediction.
Class imbalance	In a balanced dataset for binary classification problems, the number of positive and negative examples are approximately equal. Class imbalance occurs if either label occurs more frequently than the other.
Control repertoires	The comparison group that is devoid of the outcome of interest ⁹³ . Here, we refer to the comparison group that is free of disease or the immune state of interest.
Covariates	These are additional information that are collected about the individuals or samples in a study, which might influence the outcome or the patterns observed. Examples include age, gender, or genetic information (like HLA type). In ML, these are often used to help the model make more accurate predictions or to understand if other factors are affecting the results.
Cross-validation	This is a technique used in ML to evaluate how well a model will generalize to new, unseen data. The dataset is divided into multiple subsets; the model is trained on some subsets and tested on others, and this process is repeated multiple times with different subsets to get a more robust estimate of performance. It also helps in selecting the best model settings (hyperparameters).
Data leakage	This is a problem in ML where information from the test set (the data the model hasn't seen yet) unintentionally "leaks" and is accessible during training. When this happens, the model might appear to perform very well during testing, but it's actually just "cheating" by having seen some of the answers beforehand, leading to an overly optimistic and unreliable assessment of its real-world performance.
Dataset size (related terms in the literature: sample size)	The number of observations/examples available for training a machine learning model. Note that this includes both positive and negative examples. Although sample size is a commonly used term, we chose to use dataset size to avoid ambiguity with the "sample size" that can also refer to the number of clonotypes per individual in some studies.
Examples	In ML literature, "examples" typically refer to individual data

	points or observations.
Feature encoding/representation	This is the process of transforming raw data into a numerical format that a ML model can understand and process. For example, converting amino acid sequences into lists of 4-mer frequencies is a form of feature encoding.
Generalize (Generalization)	In ML, the ability of a model to "generalize" refers to how well it can accurately make predictions or perform its task on new, unseen data that was not part of its training. A good model doesn't just memorize its training data; it learns underlying patterns that apply broadly.
Ground truth	Signals introduced through simulations, where a variety of knowledge and precise control exists over the introduced signals from the outset ⁴⁹ .
Hyperparameters	These are settings or configurations of a machine learning model that are set before the training process begins (unlike the model's internal parameters, which are learned during training). Examples (in the context of manuscript's Pilot data) include the regularization constant in logistic regression or the p-value threshold in the Emerson method. Choosing the right hyperparameters is crucial for good model performance.
Immune event	Disease, allergy, vaccination, or other situations that elicit an adaptive immune response.
Immune state, immune status	Status of an individual indicating whether the adaptive immune system responded to an immune event (ongoing or past)
Immune signal, immune state-associated features, antigen-specific signal, sequence patterns (often used synonymously in the literature)	Immune memory of past and ongoing immune states, i.e. different disease or infection states stored in receptor repertoires, which may be encoded in receptor frequency, e.g., clonal expansion ³⁷ , receptor sequence, e.g., antigen-specific binding motifs or disease-associated motifs where motifs may be entire sequences ²⁵ or subsequences ^{34–36} or as a combination thereof.
Interpretability approaches	These are methods or techniques used to understand why a ML model makes certain predictions or decisions. Since complex ML models can be like "black boxes," interpretability approaches help to explain their internal workings in a way that humans can understand, especially in critical fields like medicine.
Jaccard Similarity	The Jaccard similarity ⁹⁰ measures the similarity between finite sample sets and is computed as the ratio of the size of the intersection to the size of the union.
Kaggle competition	Kaggle is an online platform that hosts ML and data science competitions. Companies and researchers pose problems, provide

	data, and often offer prizes for the best solutions developed by participants. It's a way to crowdsource solutions to complex data problems and foster innovation in the field.
Label	An annotation assigned to observations/examples that represents the desired classification by a ML method. For instance, disease-positive versus disease-negative.
Leaderboard	In ML competitions (like on Kaggle), a leaderboard is a public ranking system that shows the performance of different participants' models. It allows participants to see how their solutions compare to others, usually based on a specific scoring metric, and encourages competition.
Machine learning	Machine learning allows computer systems to learn from data without being explicitly programmed. Instead of following strict rules, ML models learn to identify patterns and make predictions or decisions by analyzing large amounts of data, much like how humans learn from experience.
Model training	This is the process of teaching a ML model to find patterns and make predictions by feeding it with data to learn from. During training, the model adjusts its internal workings based on the data it sees, aiming to become accurate at its task.
Multiple instance learning	This is a specific type of ML problem where the model is given groups of data points ("bags") with an overall label for each group, but not individual labels for each data point within the group ("instances"). The challenge is to figure out what's going on at the individual data point level when only the group outcome is known. In AIRR field, repertoire label prediction problem resembles multiple instance learning, where the immune state (e.g., health or disease) of a person's entire immune repertoire (a "bag") is often known, but not the specific antigen-binding properties of each individual receptor sequence within that repertoire (the "instances").
Naive repertoires	Collection of immune receptor sequences that have not encountered their antigen yet ²³ .
Negative controls	“Negative controls” are experimental conditions where the treatment or factor being investigated is absent, serving to confirm that any observed effects are specifically due to the treatment itself and not due to other variables. ⁹⁴
Overfitting	This is a common problem in ML where a model learns the training data too well, including its noise and random fluctuations, instead of learning the underlying general patterns. An overfit model performs very well on the data it was trained on but poorly on new, unseen data.

Pattern recognition	This is a core capability of ML, where algorithms identify regularities, trends, or relationships within data. It's about finding hidden structures or common features that can be used to categorize, predict, or understand information.
Precision	In classification, precision measures how many of the items the model predicted as positive were actually positive. It answers: "Of all the items identified as X, how many were truly X?"
Prediction	In ML, "prediction" refers to the output generated by a trained model when it's given new, unseen data. It's the model's best guess or estimation about a future outcome or a characteristic of the new data, based on what it learned from the training data.
Prediction probabilities	Instead of just giving a "yes" or "no" prediction, a ML model can often output a probability score (a number between 0 and 1). This score indicates the model's confidence in its prediction for a given outcome. For example, a probability of 0.9 might mean the model is 90% sure a person has a disease.
Recall	In classification, recall measures how many of the actual positive items the model successfully identified. It answers: "Of all the items that were truly X, how many did the model find?"
ROC AUC score	The area under the curve of the receiver operating characteristic curve ⁸⁹ . For a binary classification problem, it represents the probability that the model ranks a randomly chosen positive example higher than a randomly chosen negative example.
Seronegative	In immunology, this term refers to a person who has tested negative for specific antibodies in their blood, indicating they have not been exposed to a particular pathogen or vaccine, or their immune system hasn't produced detectable antibodies.
Seropositive	In immunology, this term refers to a person who has tested positive for specific antibodies in their blood, indicating that they have been exposed to a particular pathogen (such as a virus) or have been successfully vaccinated.
Scoring scheme	In ML competitions or evaluations, a scoring scheme is a defined set of rules and metrics used to calculate and assign a score to a model's performance. This score determines how well a model is doing and is often used to rank different solutions.
Signal abundance shift	Refers to a very specific type of distributional shift, where the way data is distributed (its patterns and characteristics) changes between the dataset used to train an ML model and the data the model encounters in the real world or in a test scenario. If there's a

	significant "shift," the model might not perform as well because the new data doesn't follow the patterns it learned during training.
Species richness (in the context of manuscript)	The number of unique V(D)J nucleotide sequences captured through sequencing in a particular individual, as a consequence of sampling depth.
Test datasets	In ML, these are separate sets of data that are used to evaluate the performance of a trained model. The model has never seen this data during its training phase, so it provides an unbiased assessment of how well the model can generalize and make predictions on new, real-world information.
Training datasets	This is the primary set of data that a ML model uses to learn patterns and relationships. The model "studies" this data to understand the problem it's trying to solve and adjusts its internal parameters based on what it observes.
V(D)J recombination	Somatic rearrangement of variable (V), diversity (D) and joining (J) regions of the genes that encode antigen receptors, leading to repertoire diversity of immunoglobulins and T cell receptors. ¹¹⁶
Witness rate, signal abundance, phenotype burden (used interchangeably in the literature)	The frequency at which immune signals occur in the positive class examples. This is often calculated as the percentage of unique sequences within a positive class example that harbor a immune signal.

References

1. Mombaerts, P. *et al.* RAG-1-deficient mice have no mature B and T lymphocytes. *Cell* **68**, 869–877 (1992).
2. Noguchi, M. *et al.* Interleukin-2 receptor gamma chain mutation results in X-linked severe combined immunodeficiency in humans. *Cell* **73**, 147–157 (1993).
3. Zinkernagel, R. M. & Doherty, P. C. Restriction of in vitro T cell-mediated cytotoxicity in lymphocytic choriomeningitis within a syngeneic or semiallogeneic system. *Nature* **248**, 701–702 (1974).
4. Zinkernagel, R. M. & Doherty, P. C. Immunological surveillance against altered self components by sensitised T lymphocytes in lymphocytic choriomeningitis. *Nature* **251**, 547–548 (1974).
5. June, C. H. & Sadelain, M. Chimeric antigen receptor therapy. *N. Engl. J. Med.* **379**, 64–73 (2018).
6. Majzner, R. G. & Mackall, C. L. Clinical lessons learned from the first leg of the CAR T cell journey. *Nat. Med.* **25**, 1341–1355 (2019).
7. Akbar, R. *et al.* Progress and challenges for the machine learning-based design of fit-for-purpose monoclonal antibodies. *MAbs* **14**, 2008790 (2022).
8. Burnet, S. F. M. *The Clonal Selection Theory of Acquired Immunity ; the Abraham Flexner Lectures of Vanderbilt University*. (Cambridge University Press, 1959).
9. Hodgkin, P. D., Heath, W. R. & Baxter, A. G. The clonal selection theory: 50 years since the revolution. *Nat. Immunol.* **8**, 1019–1026 (2007).
10. O'Donnell, T. J. *et al.* Reading the repertoire: Progress in adaptive immune receptor analysis using machine learning. *Cell Syst.* **15**, 1168–1189 (2024).
11. Rappazzo, C. G. *et al.* Defining and Studying B Cell Receptor and TCR Interactions. *J. Immunol.* **211**, 311–322

- (2023).
12. Elhanati, Y., Murugan, A., Callan, C. G., Mora, T. & Walczak, A. M. Quantifying selection in immune receptor repertoires. *Proc. Natl. Acad. Sci. U. S. A.* **111**, 9875–9880 (2014).
13. Elhanati, Y. *et al.* Inferring processes underlying B-cell repertoire diversity. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* **370**, 20140243 (2015).
14. Barennes, P. *et al.* Benchmarking of T cell receptor repertoire profiling methods reveals large systematic biases. *Nat. Biotechnol.* **39**, 236–245 (2021).
15. Greiff, V., Miho, E., Menzel, U. & Reddy, S. T. Bioinformatic and Statistical Analysis of Adaptive Immune Repertoires. *Trends Immunol.* **36**, 738–749 (2015).
16. Elhanati, Y., Sethna, Z., Callan, C. G., Jr, Mora, T. & Walczak, A. M. Predicting the spectrum of TCR repertoire sharing with a data-driven model of recombination. *Immunol. Rev.* **284**, 167–179 (2018).
17. Slabodkin, A. *et al.* Individualized VDJ recombination predisposes the available Ig sequence space. *Genome Res.* (2021) doi:10.1101/gr.275373.121.
18. Venturi, V., Price, D. A., Douek, D. C. & Davenport, M. P. The molecular basis for public T-cell responses? *Nat. Rev. Immunol.* **8**, 231–238 (2008).
19. Li, H., Ye, C., Ji, G. & Han, J. Determinants of public T cell responses. *Cell Res.* **22**, 33–42 (2012).
20. Madi, A. *et al.* T-cell receptor repertoires share a restricted set of public and abundant CDR3 sequences that are associated with self-related immunity. *Genome Res.* **24**, 1603–1612 (2014).
21. Madi, A. *et al.* T cell receptor repertoires of mice and humans are clustered in similarity networks around conserved public CDR3 sequences. *Elife* **6**, (2017).
22. Pogorelyy, M. V. *et al.* Persisting fetal clonotypes influence the structure and overlap of adult human T cell receptor repertoires. *PLoS Comput. Biol.* **13**, e1005572 (2017).
23. Greiff, V. *et al.* Systems analysis reveals high genetic and antigen-driven predetermination of antibody repertoires throughout B cell development. *Cell Rep.* **19**, 1467–1478 (2017).
24. Jouannet, C., Vantomme, H., Klatzmann, D. & Mariotti-Ferrandiz, E. Benchmarking unsupervised methods for inferring TCR specificity. *bioRxiv* 2024.10.26.620398 (2024) doi:10.1101/2024.10.26.620398.
25. Emerson, R. O. *et al.* Immunosequencing identifies signatures of cytomegalovirus exposure history and HLA-mediated effects on the T cell repertoire. *Nat. Genet.* **49**, 659–665 (2017).
26. Pogorelyy, M. V. *et al.* Method for identification of condition-associated public antigen receptor sequences. *Elife* **7**, (2018).
27. Akerman, O., Isakov, H., Levi, R., Psevkin, V. & Louzoun, Y. Counting is almost all you need. *Front. Immunol.* **13**, (2023).
28. Dash, P. *et al.* Quantifiable predictive features define epitope-specific T cell receptor repertoires. *Nature* **547**, 89–93 (2017).
29. Glanville, J. *et al.* Identifying specificity groups in the T cell receptor repertoire. *Nature* **547**, 94–98 (2017).
30. Zhang, H. *et al.* Investigation of Antigen-Specific T-Cell Receptor Clusters in Human Cancers. *Clin. Cancer Res.* **26**, 1359–1371 (2020).
31. Widrich, M. *et al.* Modern hopfield networks and attention for immune repertoire classification. *Adv. Neural Inf. Process. Syst.* **33**, 18832–18845 (2020).
32. Zaslavsky, M. E. *et al.* Disease diagnostics using machine learning of B cell and T cell receptor sequences. *Science* **387**, eadp2407 (2025).
33. Ostmeyer, J. *et al.* Biophysicochemical motifs in T cell receptor sequences as a potential biomarker for high-grade serous ovarian carcinoma. *PLoS One* **15**, e0229569 (2020).
34. Ostmeyer, J., Christley, S., Toby, I. T. & Cowell, L. G. Biophysicochemical Motifs in T-cell Receptor Sequences Distinguish Repertoires from Tumor-Infiltrating Lymphocyte and Adjacent Healthy Tissue. *Cancer Res.* **79**, 1671–1680 (2019).
35. Akbar, R. *et al.* A compact vocabulary of paratope-epitope interactions enables predictability of antibody-antigen binding. *Cell Rep.* **34**, 108856 (2021).
36. Shrock, E. L. *et al.* Germline-encoded amino acid-binding motifs drive immunodominant public antibody

- responses. *Science* **380**, eadc9498 (2023).
37. Greiff, V. *et al.* A bioinformatic framework for immune repertoire diversity profiling enables detection of immunological status. *Genome Med.* **7**, 49 (2015).
 38. Greiff, V., Yaari, G. & Cowell, L. G. Mining adaptive immune receptor repertoires for biological and clinical information using machine learning. *Current Opinion in Systems Biology* **24**, 109–119 (2020).
 39. Chernigovskaya, M. *et al.* Simulation of adaptive immune receptors and repertoires with complex immune information to guide the development and benchmarking of AIRR machine learning. *Nucleic Acids Res.* **53**, gkaf025 (2025).
 40. Bashford-Rogers, R. J. M. *et al.* Analysis of the B cell receptor repertoire in six immune-mediated diseases. *Nature* **574**, 122–126 (2019).
 41. Pertseva, M., Gao, B., Neumeier, D., Yermanos, A. & Reddy, S. T. Applications of Machine and Deep Learning in Adaptive Immunity. *Annu. Rev. Chem. Biomol. Eng.* **12**, 39–62 (2021).
 42. Mhanna, V. *et al.* Adaptive immune receptor repertoire analysis. *Nature Reviews Methods Primers* **4**, 1–25 (2024).
 43. Sidhom, J.-W. *et al.* Deep learning reveals predictive sequence concepts within immune repertoires to immunotherapy. *Sci Adv* **8**, eabq5089 (2022).
 44. Mora, T. & Walczak, A. M. How many different clonotypes do immune repertoires contain? *Curr. Opin. Syst. Biol.* **18**, 104–110 (2019).
 45. Dietterich, T. G., Lathrop, R. H. & Lozano-Pérez, T. Solving the multiple instance problem with axis-parallel rectangles. *Artif. Intell.* **89**, 31–71 (1997).
 46. Maron, O. & Lozano-Pérez, T. A framework for multiple-instance learning. in *Proceedings of the 1997 conference on Advances in neural information processing systems 10* 570–576 (MIT Press, Cambridge, MA, USA, 1998).
 47. Wang, X., Yan, Y., Tang, P., Bai, X. & Liu, W. Revisiting multiple instance neural networks. *Pattern Recognit.* **74**, 15–24 (2018).
 48. Arnaout, R. A., Prak, E. T. L., Schwab, N., Rubelt, F. & Adaptive Immune Receptor Repertoire Community. The Future of Blood Testing Is the Immunome. *Front. Immunol.* **12**, 626793 (2021).
 49. Sandve, G. K. & Greiff, V. Access to ground truth at unconstrained size makes simulated data as indispensable as experimental data for bioinformatics methods development and benchmarking. *Bioinformatics* (2022) doi:10.1093/bioinformatics/btac612.
 50. Murphy, K. P. *Probabilistic Machine Learning: An Introduction*. (MIT Press, 2022).
 51. Weber, L. M. *et al.* Essential guidelines for computational method benchmarking. *Genome Biol.* **20**, 125 (2019).
 52. Chen, V. *et al.* Best Practices for Interpretable Machine Learning in Computational Biology. *bioRxiv* 2022.10.28.513978 (2022) doi:10.1101/2022.10.28.513978.
 53. Moris, P. *et al.* Current challenges for unseen-epitope TCR interaction prediction and a new perspective derived from image classification. *Brief. Bioinform.* **22**, (2021).
 54. Pavlović, M. *et al.* Improving generalization of machine learning-identified biomarkers with causal modeling: an investigation into immune receptor diagnostics. *arXiv [q-bio.QM]* (2022).
 55. Pavlović, M. *et al.* The immuneML ecosystem for machine learning analysis of adaptive immune receptor repertoires. *Nature Machine Intelligence* **3**, 936–944 (2021).
 56. Akbar, R. *et al.* In silico proof of principle of machine learning-based antibody design at unconstrained scale. *MAbs* **14**, 2031482 (2022).
 57. Robert, P. A. *et al.* Unconstrained generation of synthetic antibody–antigen structures to guide machine learning methodology for antibody specificity prediction. *Nature Computational Science* **2**, 845–865 (2022).
 58. Marcou, Q., Mora, T. & Walczak, A. M. High-throughput immune repertoire analysis with IGoR. *Nat. Commun.* **9**, 561 (2018).
 59. Weber, C. R. *et al.* immuneSIM: tunable multi-feature simulation of B- and T-cell receptor repertoires for immunoinformatics benchmarking. *Bioinformatics* (2020) doi:10.1093/bioinformatics/btaa158.

60. Woodcock, M. G., Bortone, D. S. & Vincent, B. G. STIG: Generation and simulated sequencing of synthetic T cell receptor repertoires. *bioRxiv* 2020.02.28.969469 (2020) doi:10.1101/2020.02.28.969469.
61. Safonova, Y., Lapidus, A. & Lill, J. IgSimulator: a versatile immunosequencing simulator. *Bioinformatics* **31**, 326–332 (2015).
62. Yermanos, A. *et al.* Comparison of methods for phylogenetic B-cell lineage inference using time-resolved antibody repertoire simulations (AbSim). *Bioinformatics* (2017) doi:10.1093/bioinformatics/btx533.
63. Han, J. *et al.* Echidna: Integrated simulations of single-cell immune receptor repertoires and transcriptomes. *Bioinformatics Advances* **2**, vbac062 (2022).
64. Sutherland, C. & Cowan, G. J. M. AIRRSHIP: simulating human B cell receptor repertoire sequences. Preprint at <https://doi.org/10.1101/2022.12.20.521228>.
65. Konstantinovskiy, T. & Yaari, G. A novel approach to T-cell receptor beta chain (TCRB) repertoire encoding using lossless string compression. *Bioinformatics* **39**, (2023).
66. Kanduri, C. *et al.* simAIRR: simulation of adaptive immune repertoires with realistic receptor sequence sharing for benchmarking of immune state prediction methods. *Gigascience* **12**, (2022).
67. Chernigovskaya, M. *et al.* Simulation of adaptive immune receptors and repertoires with complex immune information to guide the development and benchmarking of AIRR machine learning. Preprint at <https://doi.org/10.1101/2023.10.20.562936> (2023).
68. Kanduri, C. *et al.* Profiling the baseline performance and limits of machine learning models for adaptive immune receptor repertoire classification. *Gigascience* **11**, (2022).
69. Rawat, P. *et al.* Identification of a type 1 diabetes-associated T cell receptor repertoire signature from the human peripheral blood. *medRxiv* 2024.12.10.24318751 (2024).
70. Nolan, S. *et al.* A large-scale database of T-cell receptor beta (TCR β) sequences and binding associations from natural and synthetic exposure to SARS-CoV-2. *Res Sq* (2020) doi:10.21203/rs.3.rs-51964/v1.
71. Russakovsky, O. *et al.* ImageNet Large Scale Visual Recognition Challenge. *arXiv [cs.CV]* (2014).
72. Bennett, J. & Lanning, S. The Netflix prize. *Proceedings of KDD cup and workshop* (2007).
73. LeCun, Y. The MNIST database of handwritten digits. <http://yann.lecun.com/exdb/mnist/> (1998).
74. Kryshtafovych, A., Schwede, T., Topf, M., Fidelis, K. & Moult, J. Critical assessment of methods of protein structure prediction (CASP)-Round XV. *Proteins* **91**, 1539–1549 (2023).
75. Stolovitzky, G., Monroe, D. & Califano, A. Dialogue on reverse-engineering assessment and methods: the DREAM of high-throughput pathway inference: The DREAM of high-throughput pathway inference. *Ann. N. Y. Acad. Sci.* **1115**, 1–22 (2007).
76. Nielsen, M. *et al.* Lessons learned from the IMMREP23 TCR-epitope prediction challenge. *Immunoinformatics (Amst.)* **16**, 100045 (2024).
77. Erasmus, M. F. *et al.* AIntibody: an experimentally validated in silico antibody discovery design challenge. *Nat. Biotechnol.* **42**, 1637–1642 (2024).
78. Kaggle: Your Machine Learning and Data Science Community. <https://www.kaggle.com/>.
79. Christophersen, A. *et al.* Tetramer-visualized gluten-specific CD4⁺ T cells in blood as a potential diagnostic marker for coeliac disease without oral gluten challenge. *United European Gastroenterol J* **2**, 268–278 (2014).
80. Weber, C. R. *et al.* Reference-based comparison of adaptive immune receptor repertoires. *Cell Rep Methods* **2**, 100269 (2022).
81. Jabran Zahid, H. *et al.* Large-scale statistical mapping of T-cell receptor β sequences to Human Leukocyte Antigens. *bioRxiv* 2024.04.01.587617 (2024) doi:10.1101/2024.04.01.587617.
82. Rubelt, F. *et al.* Adaptive Immune Receptor Repertoire Community recommendations for sharing immune-repertoire sequencing data. *Nat. Immunol.* **18**, 1274–1278 (2017).
83. Vander Heiden, J. A. *et al.* AIRR Community standardized representations for annotated immune repertoires. *Front. Immunol.* **9**, 2206 (2018).
84. Cinelli, M. *et al.* Feature selection using a one dimensional naïve Bayes' classifier increases the accuracy of support vector machine classification of CDR3 repertoires. *Bioinformatics* **33**, 951–955 (2017).
85. Sun, Y. *et al.* Specificity, Privacy, and Degeneracy in the CD4 T Cell Receptor Repertoire Following

- Immunization. *Front. Immunol.* **8**, 430 (2017).
86. Thomas, N. *et al.* Tracking global changes induced in the CD4 T-cell receptor repertoire by immunization with a complex antigen using short stretches of CDR3 protein sequence. *Bioinformatics* **30**, 3181–3188 (2014).
87. Katayama, Y. & Kobayashi, T. J. Comparative Study of Repertoire Classification Methods Reveals Data Efficiency of k -mer Feature Extraction. *Front. Immunol.* **13**, 797640 (2022).
88. Sethna, Z., Elhanati, Y., Callan, C. G., Walczak, A. M. & Mora, T. OLGA: fast computation of generation probabilities of B- and T-cell receptor amino acid sequences and motifs. *Bioinformatics* **35**, 2974–2981 (2019).
89. Bradley, A. P. The use of the area under the ROC curve in the evaluation of machine learning algorithms. *Pattern Recognit.* **30**, 1145–1159 (1997).
90. Costa, L. da F. Further generalizations of the Jaccard index. *arXiv [cs.LG]* (2021).
91. Corrie, B. D. *et al.* iReceptor: A platform for querying and analyzing antibody/B-cell and T-cell receptor repertoire data across federated repositories. *Immunol. Rev.* **284**, 24–41 (2018).
92. Christley, S. *et al.* The ADC API: A web API for the programmatic query of the AIRR Data Commons. *Front. Big Data* **3**, 22 (2020).
93. Grimes, D. A. & Schulz, K. F. Compared to what? Finding controls for case-control studies. *Lancet* **365**, 1429–1433 (2005).
94. Lipsitch, M., Tchetgen Tchetgen, E. & Cohen, T. Negative controls: a tool for detecting confounding and bias in observational studies. *Epidemiology* **21**, 383–388 (2010).
95. Bagaev, D. V. *et al.* VDJdb in 2019: database extension, new analysis infrastructure and a T-cell receptor motif compendium. *Nucleic Acids Res.* **48**, D1057–D1062 (2020).
96. AIRR Community. Rearrangement Schema — AIRR Standards 1.4 documentation. <https://docs.airr-community.org/en/latest/datarep/rearrangements.html>.
97. Widrich, M. *et al.* Modern Hopfield Networks and Attention for Immune Repertoire Classification. *arXiv [cs.LG]* (2020).
98. Sikt FileSender. <https://filesender.sikt.no/>.
99. Slack. Where work happens. <https://slack.com/intl/en-gb/>.
100. Pourghesari, B. *et al.* The cytomegalovirus-specific CD4+ T-cell response expands with age and markedly alters the CD4+ T-cell repertoire. *J. Virol.* **81**, 7759–7765 (2007).
101. Marquez, S. *et al.* Adaptive immune receptor repertoire (AIRR) community guide to repertoire analysis. *Methods Mol. Biol.* **2453**, 297–316 (2022).
102. Torralba, A. & Efros, A. A. Unbiased look at dataset bias. *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.* 1521–1528 (2011).
103. Daume, H., III & Marcu, D. Domain adaptation for statistical classifiers. *J. Artif. Intell. Res.* **26**, 101–126 (2006).
104. Dataset Shift in Machine Learning. *MIT Press* <https://mitpress.mit.edu/9780262545877/dataset-shift-in-machine-learning/> (2021).
105. He, H. & Garcia, E. A. Learning from imbalanced data. *IEEE Trans. Knowl. Data Eng.* **21**, 1263–1284 (2009).
106. Krawczyk, B. Learning from imbalanced data: open challenges and future directions. *Prog. Artif. Intell.* **5**, 221–232 (2016).
107. Buda, M., Maki, A. & Mazurowski, M. A. A systematic study of the class imbalance problem in convolutional neural networks. *Neural Netw.* **106**, 249–259 (2018).
108. Pavlović, M. *et al.* Improving generalization of machine learning-identified biomarkers using causal modelling with examples from immune receptor diagnostics. *Nature Machine Intelligence* **6**, 15–24 (2024).
109. Boulesteix, A.-L., Binder, H., Abrahamowicz, M., Sauerbrei, W. & Simulation Panel of the STRATOS Initiative. On the necessity and design of studies comparing statistical methods. *Biom. J.* **60**, 216–218 (2018).
110. Boulesteix, A.-L., Lauer, S. & Eugster, M. J. A. A plea for neutral comparison studies in computational sciences. *PLoS One* **8**, e61562 (2013).
111. Buchka, S., Hapfelmeier, A., Gardner, P. P., Wilson, R. & Boulesteix, A.-L. On the optimistic performance evaluation of newly introduced bioinformatic methods. *Genome Biol.* **22**, 152 (2021).

112. Norel, R., Rice, J. J. & Stolovitzky, G. The self-assessment trap: can we all be better than average? *Mol. Syst. Biol.* **7**, 537 (2011).
113. Mangul, S. *et al.* Systematic benchmarking of omics computational tools. *Nat. Commun.* **10**, 1393 (2019).
114. Brooks, T. G., Lahens, N. F., Mrčela, A. & Grant, G. R. Challenges and best practices in omics benchmarking. *Nat. Rev. Genet.* **25**, 326–339 (2024).
115. Megahed, F. M. *et al.* Comparing classifier performance with baselines. *Nat. Methods* **21**, 546–548 (2024).
116. Schatz, D. G. & Ji, Y. Recombination centres and the orchestration of V(D)J recombination. *Nat. Rev. Immunol.* **11**, 251–263 (2011).

Acknowledgements

Maria Mamica and Charlotte Wurtzen are thanked for brainstorming on the illustrations shown in Figures 2 and 3. A large majority of the computations in this work were performed using the Immunohub eInfrastructure funded by the University of Oslo and operated by the authors in close collaboration with the University Center for Information Technology (USIT), University of Oslo.

Author contributions

Author contributions are categorized according to Contributor Role Taxonomy (CRediT; <https://credit.niso.org>).

- Conceptualization: CK, GKS, VG, GY
- Data Curation: CK (major); PR (involved with initial preprocessing of T1D cohort)
- Formal Analysis: CK
- Funding Acquisition: GKS, VG, GY
- Investigation: CK
- Methodology: CK, GKS, VG, GY (major); TK (idea related to Kaggle platform)
- Project Administration: CK, GKS, VG
- Software: CK (major); MP (adapted code for generation of synthetic datasets that assume shared sequence patterns as immune signals)
- Resources: GKS, VG (related to computing resources); TMB (related to T1D cohort), DHM, RE, BHo, HSR (team from Adaptive Biotechnologies related to HSV-2 cohort), CC, BHa, AK, JJ, EP, CR, AR (team from Parse Biosciences related to experimental antigen-specific sequences)
- Supervision: GKS, VG, GY
- Validation: CK (major); TK, PR (initial internal assessment of Kaggle datasets)
- Visualization: CK
- Writing - Original Draft: CK
- Writing - Review & Editing: All authors, with particular emphasis on JB, GK, EMF, PM, ELP, LGC, TMB, and AIRR-C ML working group for their critical conceptual feedback.

Competing interests

V.G. declares advisory board positions in aiNET GmbH, Enpicom B.V, Specifica Inc, Adaptyv Biosystems, EVQLV, Omniscope, Diagonal Therapeutics, and Absci. V.G. is a consultant for Roche/Genentech, immunai, Proteinea, and LabGenius. LGC is the chief scientific officer for OncoSeer Diagnostics. PM holds shares in ImmuneWatch BV, an immunoinformatics company. JB is an employee of Xaira Therapeutics. All the remaining authors declare no competing interests. DHM, RE, BHo, HSR are employees of Adaptive Biotechnologies. CC, BHa, AK, JJ, EP, CR, AR are employees of Parse Biosciences. P.R. serves as an advisor to The(sugar)science for the DChallenge 2025 initiative focused on Type 1 Diabetes (T1D).

Funding information

We acknowledge generous support by The Leona M. and Harry B. Helmsley Charitable Trust (#2019PG-T1D011, to VG and TMB), UiO World-Leading Research Community (to VG), UiO:LifeScience Convergence Environment Immunolingo (to VG, GKS), EU Horizon 2020 iReceptorplus (#825821) (to VG), Research Council of Norway projects (#300740, 331890 to VG), a Research Council of Norway IKTPLUSS project (#311341, to VG and GKS), a Norwegian Cancer Society Grant (#215817, to VG), and Stiftelsen Kristian Gerhard Jebsen (K.G. Jebsen Coeliac Disease Research Centre) (to GKS), UiO:LifeScience Convergence Environment RealArt (to GKS, CK), the National Institutes of Health (P01 AI042288, to TMB), U24AI177622 to LGC, ANR-23-CE17-0039-01 to EMF, NIH 2U0-DK112217, U19AI110483, P30CA016520 to ELP, EU Horizon 2020 research and innovation Marie Skłodowska-Curie grant 801133 (to PR). This project has received funding from the Innovative Medicines Initiative (IMI) 2 Joint Undertaking under grant agreement No 101007799 (Inno4Vac). This Joint Undertaking receives support from the European Union's Horizon 2020 research and innovation programme and EFPIA (to VG). This communication reflects the author's view and neither IMI nor the European Union, EFPIA, or any Associated Partners are responsible for any use that may be made of the information contained therein. This work was carried out on immunohub e-Infrastructure funded by the University of Oslo and jointly operated by GreiffLab and SandveLab (the authors) in close collaboration with the University Center for Information Technology, University of Oslo, IT-Department (USIT). This study was partially supported by grants from the ISF (2940/21), VATAT, and Ministry of Innovation, Science & Technology (1001576181/0004941) (to GY).

Table S4: Description of Phase-2 datasets with increasing signal abundances and diverse immune signal patterns (related to **Q1 in Box 2**)

Datasets	Type of dataset	Immune state -associated features	Background sequences used for repertoire construction	Number of witnesses per repertoire in positive labelled repertoires	Total number of immune state-associated witnesses in the dataset	Number of independent replicated datasets
Q1-1	Experimental TCR beta CMV+ versus CMV-	Not Applicable	Not Applicable	Not Applicable	Not Applicable	1
Q1-2	Experimental TCR beta T1D+ versus T1D-	Not Applicable	Not Applicable	Not Applicable	Not Applicable	1
Q1-3-5	Synthetic	GAD65, EBV, MART1 specific sequences acquired from Parse Biosciences	A random sample of experimental TCR sequences from ImmuneCode database	5	~ 650	3
Q1-6-8	Synthetic	Proinsulin, EBV, MART1 specific sequences acquired from Parse Biosciences	A random sample of experimental TCR sequences from ImmuneCode database	20	~ 750-1020	3
Q1-9-11	Synthetic	GAD65, EBV, MART1 specific sequences acquired from Parse Biosciences	A random sample of experimental TCR sequences from ImmuneCode database	100	~ 1615-1710	3
Q1-12-14	Synthetic	Sequences sharing short sub-sequence patterns spanning 4 to 6 amino acid residues with up to a maximum of 3 gapped positions	A random sample of synthetic TCR sequences drawn from known V(D)J recombination models	5	~ 2000	3
Q1-15-17	Synthetic	Sequences sharing short sub-sequence patterns spanning 4 to 6 amino acid residues with up to a maximum of 3 gapped positions	A random sample of synthetic TCR sequences drawn from known V(D)J recombination models	20	~ 8000	3
Q1-18-20	Synthetic	Sequences sharing short sub-sequence patterns spanning 4 to 6 amino acid residues with up to a maximum of 3 gapped positions	A random sample of synthetic TCR sequences drawn from known V(D)J recombination models	100	~ 40000	3

Table S5: Description of phase-2 datasets with varying dataset sizes (related to **Q2 in Box 2**)

Datasets	Type of dataset	Dataset size with balanced labels	Immune state - associated features	Background sequences used for repertoire construction	Number of witnesses per repertoire in positive labelled repertoires	Total number of immune state-associated witnesses in the dataset	Number of independent replicated datasets
Q2-1-3	Experimental TCR beta T1D+ versus T1D-	100	Not Applicable	Not Applicable	Not Applicable	Not Applicable	3
Q2-4-6	Experimental TCR beta T1D+ versus T1D-	200	Not Applicable	Not Applicable	Not Applicable	Not Applicable	3
Q2-7-9	Experimental TCR beta T1D+ versus T1D-	400	Not Applicable	Not Applicable	Not Applicable	Not Applicable	3
Q2-10-12	Experimental TCR beta CMV+ versus CMV-	100	Not Applicable	Not Applicable	Not Applicable	Not Applicable	3
Q2-13-15	Experimental TCR beta CMV+ versus CMV-	200	Not Applicable	Not Applicable	Not Applicable	Not Applicable	3
Q2-16-18	Experimental TCR beta CMV+ versus CMV-	400	Not Applicable	Not Applicable	Not Applicable	Not Applicable	3
Q2-19-21	Synthetic	100	MBP, HPV, MART1 specific sequences acquired from Parse Biosciences	A random sample of experimental TCR sequences from ImmuneCode database	20	~ 920-1030	3
Q2-22-24	Synthetic	200	MBP, HPV, MART1 specific sequences acquired from Parse Biosciences	A random sample of experimental TCR sequences from ImmuneCode database	20	~ 920-1030	3
Q2-25-27	Synthetic	400	Proinsulin, EBV, MART1 specific sequences acquired from Parse Biosciences	A random sample of experimental TCR sequences from ImmuneCode database	20	~ 750-1020	3

Table S6: Description of Phase-2 datasets with varying Species richness (related to **Q3 in Box 2**)

Datasets	Type of dataset	Species richness	Immune state - associated features	Background sequences used for repertoire construction	Number of witnesses per repertoire in positive-labelled repertoires	Total number of immune state-associated witnesses in the dataset	Number of independent replicated datasets
Q3-1-3	Synthetic	6k sequences	Influenza A-specific sequences downloaded from VDJdb	A random sample of experimental TCR sequences from ImmuneCode database	10	~ 330	3
Q3-4-6	Synthetic	12k sequences	Influenza A-specific sequences downloaded from VDJdb	A random sample of experimental TCR sequences from ImmuneCode database	20	~ 650	3
Q3-7-9	Synthetic	25k sequences	Influenza A-specific sequences downloaded from VDJdb	A random sample of experimental TCR sequences from ImmuneCode database	40	~ 880	3
Q3-10-12	Experimental TCR beta T1D+ versus T1D-	Subsampled to 6k sequences	Not Applicable	Not Applicable	Not Applicable	Not Applicable	3
Q3-13-15	Experimental TCR beta T1D+ versus T1D-	Subsampled to 12k sequences	Not Applicable	Not Applicable	Not Applicable	Not Applicable	3
Q3-16-18	Experimental TCR beta T1D+ versus T1D-	Subsampled to 25k sequences	Not Applicable	Not Applicable	Not Applicable	Not Applicable	3
Q3-19-21	Experimental TCR beta CMV+ versus CMV-	Subsampled to 6k sequences	Not Applicable	Not Applicable	Not Applicable	Not Applicable	3
Q3-22-24	Experimental TCR beta CMV+ versus CMV-	Subsampled to 12k sequences	Not Applicable	Not Applicable	Not Applicable	Not Applicable	3
Q3-25-27	Experimental TCR beta CMV+ versus CMV-	Subsampled to 25k sequences	Not Applicable	Not Applicable	Not Applicable	Not Applicable	3

Table S7: Description of phase-2 datasets with varying signal abundance shifts (related to **Q4 in Box 2**)

Datasets	Type of dataset	Immune state - associated features	Background sequences used for repertoire construction	Number of witnesses per repertoire in positive-labeled repertoires in training data	Number of witnesses per repertoire in positive-labeled repertoires in test data	Total number of immune state-associated witnesses in the dataset	Number of independent replicated datasets
Q4-1-3	Synthetic	EBV-specific sequences downloaded from VDJdb	A random sample of experimental TCR sequences from ImmuneCode database	40	20	~ 870	3
Q4-4-6	Synthetic	EBV-specific sequences downloaded from VDJdb	A random sample of experimental TCR sequences from ImmuneCode database	20	5	~ 650	3

Table S8: Description of Phase-2 datasets with varying class imbalances (related to **Q5 in Box 2**)

Datasets	Type of dataset	Immune state - associated features	Background sequences used for repertoire construction	Number of witnesses per repertoire in positive-labeled repertoires	Percentage of positive examples in training data	Total number of immune state-associated witnesses in the dataset	Number of independent replicated datasets
Q5-1-3	Synthetic	CMV-specific sequences downloaded from VDJdb	A random sample of experimental TCR sequences from ImmuneCode database	20	20%	~ 650	3
Q5-4-6	Synthetic	CMV-specific sequences downloaded from VDJdb	A random sample of experimental TCR sequences from ImmuneCode database	20	80%	~ 650	3

Table S9: Description of phase-2 datasets to investigate the impact of systematic differences between comparator groups (e.g., with distinct experimental protocols and other batch effects etc; related to **Q6 in Box 2**)

Dataset	Type of dataset	Train	Test	Training dataset size	Test dataset size	Training dataset protocols (see footer for codes)	Test dataset protocols (see footer for codes)	Number of replications
Q6-1	Experimental TRB	T1D+/T1D-	T1D+/T1D-	400	200	Positive class: 1 Negative class: 1	Positive class: 2 Negative class: 2	1
Q6-2	Experimental TRB	T1D+/CMV-	T1D+/CMV-	400	200	Positive class: 1 Negative class: 3,4	Positive class: 1 Negative class: 3,4	1
Q6-3	Experimental TRB	T1D+/CMV-	T1D+/T1D-	400	200	Positive class: 1 Negative class: 3,4	Positive class: 2 Negative class: 3,4	1
Q6-4	Experimental TRB	T1D+/SARS-CoV-2+	T1D+/SARS-CoV-2+	400	200	Positive class: 1 Negative class: 1	Positive class: 1 Negative class: 1	1
Q6-5	Experimental TRB	T1D+/SARS-CoV-2+	T1D+/T1D-	400	200	Positive class: 1 Negative class: 1	Positive class: 2 Negative class: 2	1
Q6-6	Experimental TRB	CMV+/CMV-	CMV+/CMV-	400	200	Positive class: 3,4 Negative class: 3,4	Positive class: 3,4 Negative class: 3,4	1
Q6-7	Experimental TRB	CMV+/T1D-	CMV+/T1D-	400	200	Positive class: 3,4 Negative class: 1	Positive class: 3,4 Negative class: 2	1
Q6-8	Experimental TRB	CMV+/T1D-	CMV+/CMV-	400	200	Positive class: 3,4 Negative class: 1	Positive class: 3,4 Negative class: 3,4	1
Q6-9	Experimental TRB	CMV+/SARS-CoV-2+	CMV+/SARS-CoV-2+	400	200	Positive class: 3,4 Negative class: 1	Positive class: 3,4 Negative class: 1	1
Q6-10	Experimental TRB	CMV+/SARS-CoV-2+	CMV+/CMV-	400	200	Positive class: 3,4 Negative class: 1	Positive class: 3,4 Negative class: 3,4	1

1=Adaptive Biotechnologies immunoSEQ assay version hsTCRB_v4; 2=Adaptive Biotechnologies immunoSEQ assay version hsTCRB_v4b; 3=Adaptive Biotechnologies immunoSEQ assay version hsTCRB_v2; 4=Adaptive Biotechnologies immunoSEQ assay version hsTCRB_v3. 1,2 were used for T1D; 3,4 for CMV; 1 for SARS-CoV-2 repertoires.

Table S10: Description of Phase-2 datasets to investigate if methods are able to not report optimistic findings when there is no signal (negative control datasets; related to **Q7 in Box 2**)

Datasets	Type of dataset	Immune state - associated features	Background sequences used for repertoire construction	Number of witnesses per repertoire in positive-labelled repertoires	Number of independent replicated datasets
Q7-1	Experimental TRB T1D+ vs T1D- with permuted labels	None	Not applicable	Not applicable	1
Q7-2	Synthetic	None	A random sample of experimental TCR sequences from ImmuneCode database	0	1